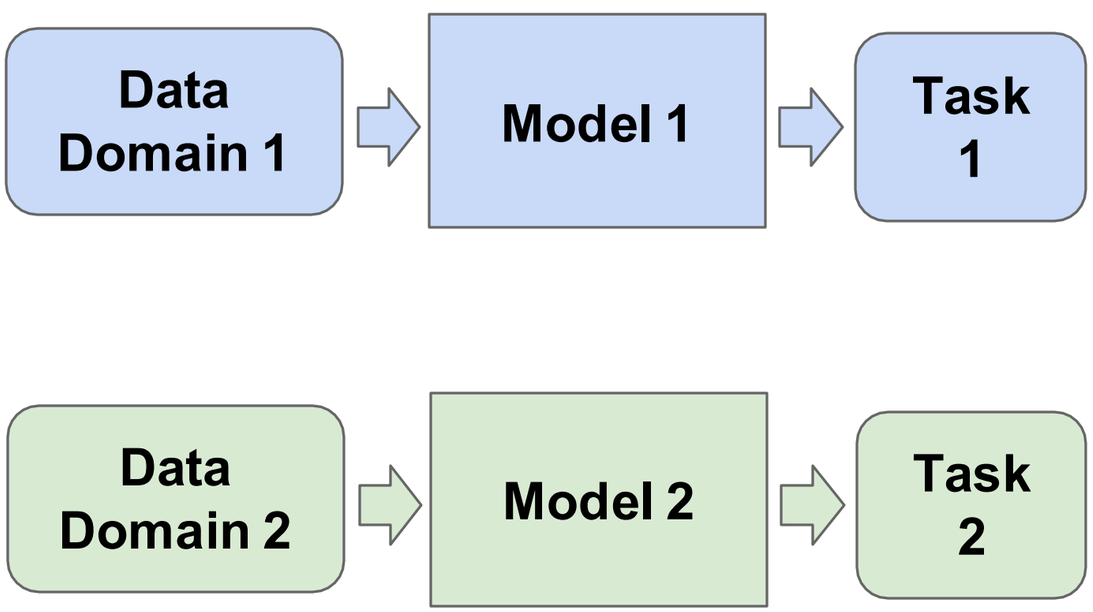
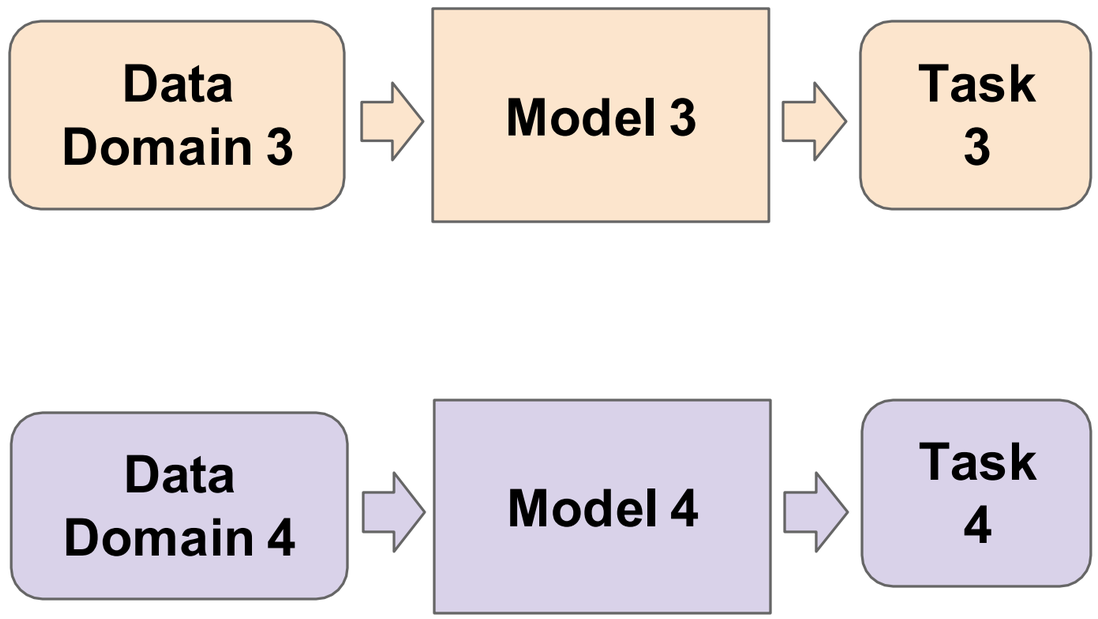
Vision-Language Foundation Models

How have we been thinking about models in this class so far?

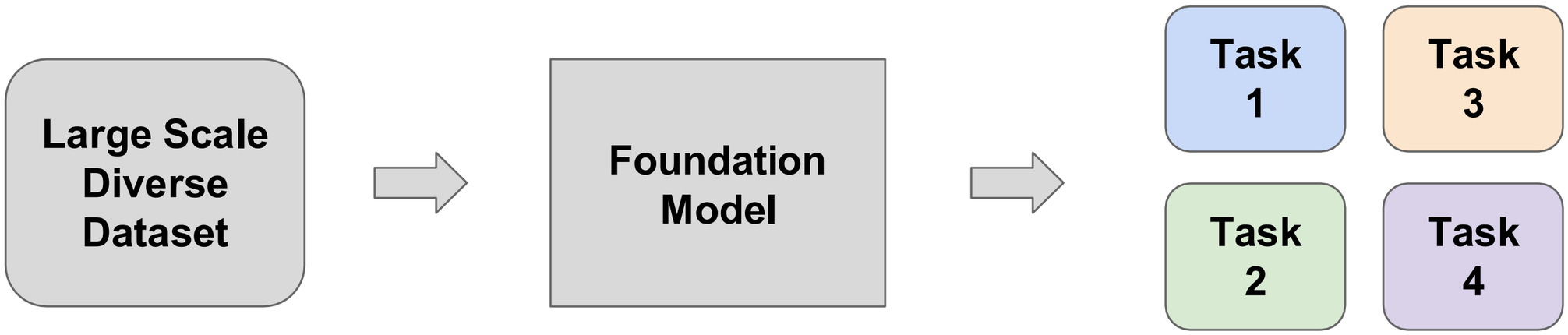
Train a *specialized* model for *each* task





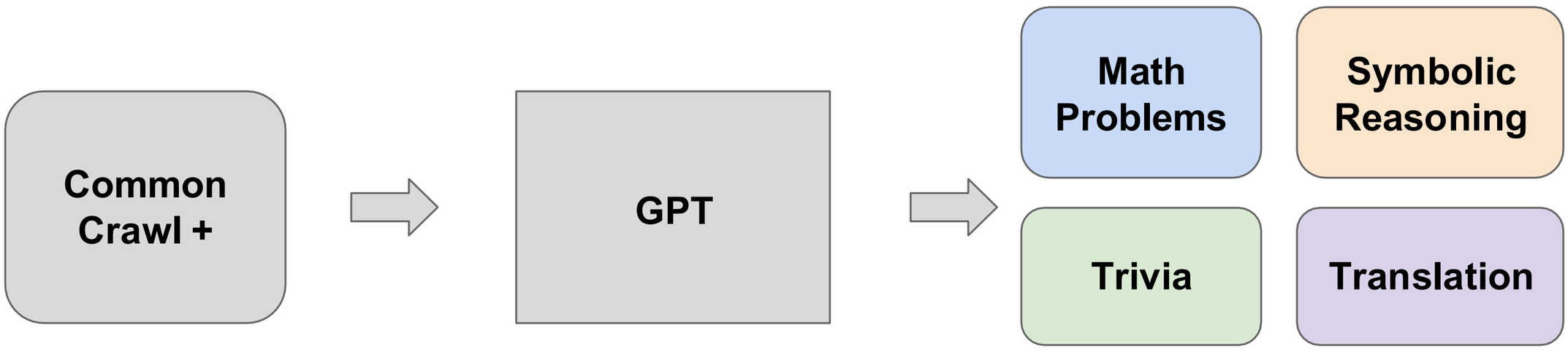
Now, we build Foundation Models

Pre-train one model that acts as the *foundation* for many different tasks



Foundation Models

Language



There are many classes of Foundation Models

Language	Classification	LM + Vision	And More!	Chaining
ELMo	CLIP	LLaVA	Segment Anything	LMs + CLIP
BERT	CoCa	Flamingo	Whisper	Visual Programming
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

How do identify a model as a Foundation?

Always see with foundation models:

- general /robust to many different tasks

Often see with foundation models:

- Large # params
- Large amount of data
- Self-supervised pre-training objective

Language models are out of scope for this class

Language	Classification	LM + Vision	And More!	Chaining
ELMo	CLIP	LLaVA	Segment Anything	LMs + CLIP
BERT	CoCa	Flamingo	Whisper	Visual Programming
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

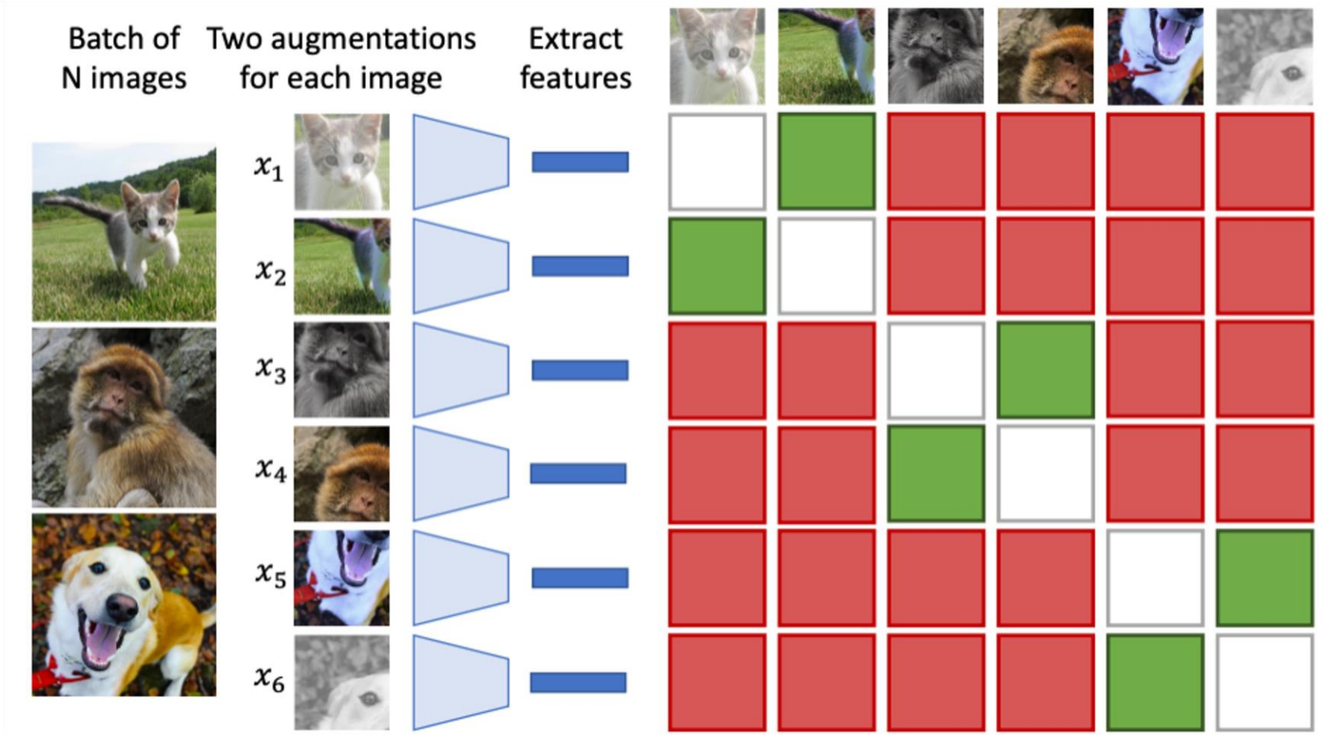
We will focus on multimodal (vision) foundation models

Language	Classification	LM + Vision	And More!	Chaining
ELMo	**CLIP**	**LLaVA**	**Segment Anything**	**LMs + CLIP**
BERT	CoCa	Flamingo	Whisper	**Visual Programming**
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

Let’s start with the foundation models for classification

Language	Classification	LM + Vision	And More!	Chaining
ELMo	**CLIP**	**LLaVA**	**Segment Anything**	**LMs + CLIP**
BERT	CoCa	Flamingo	Whisper	**Visual Programming**
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

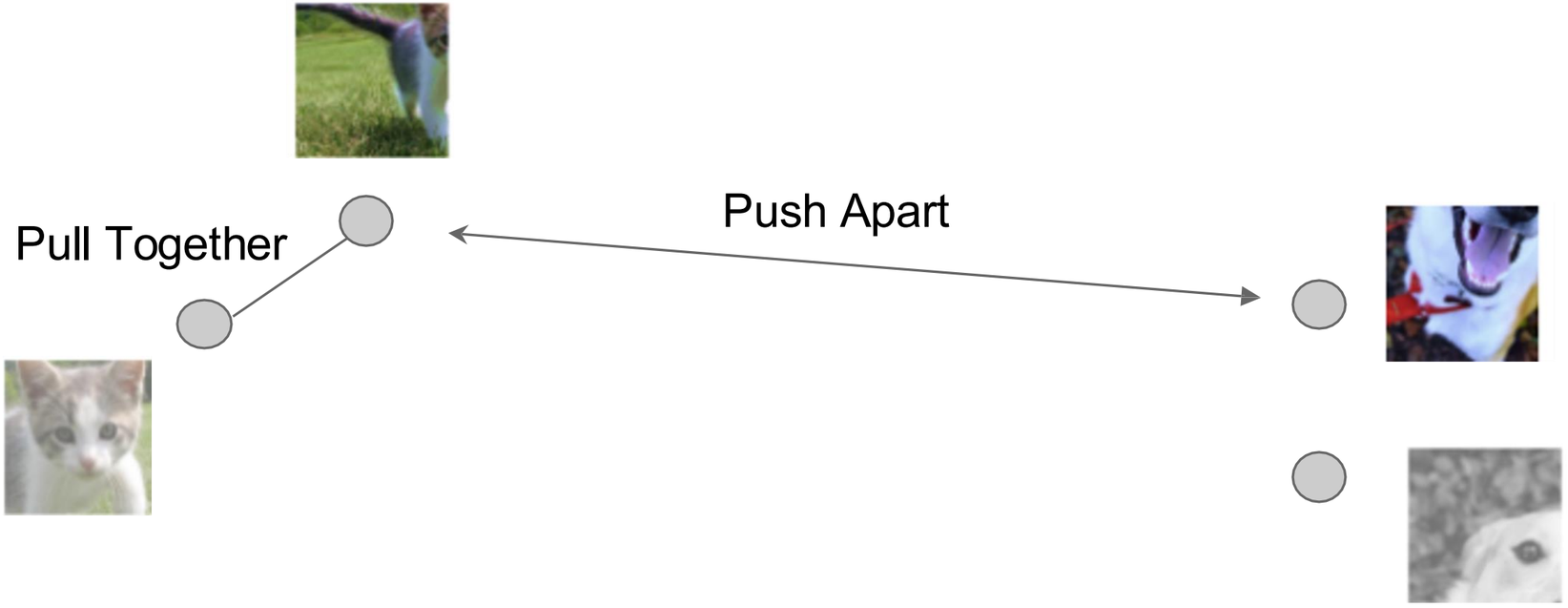
Recall this self-supervised objective from SimCLR



Use Self Supervised learning to learn good image features

Can train small classifiers on top of these features using supervised learning

The main idea was to learning concepts without labels -> a self-supervised pretraining objective



The hope was that the learned representations generalize to new instances









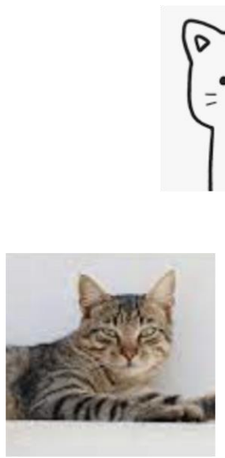








Can we generalize these representations beyond just images? *To language perhaps?*







B





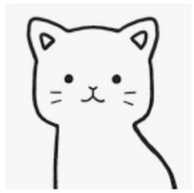




A

1. “A cute fluffy cat”
2. “My favorite dog is a golden retriever”

What if this representation space could also embed sentences/phrases?









“A cute fluffy cat”

“My favorite dog is a golden retriever”

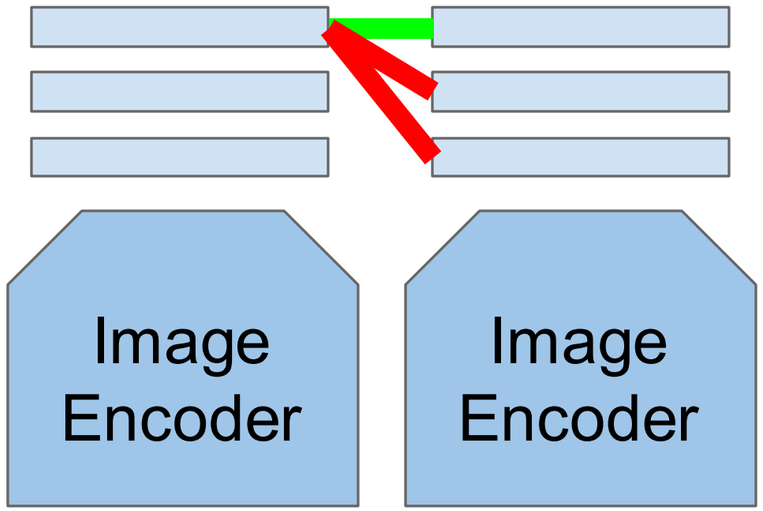


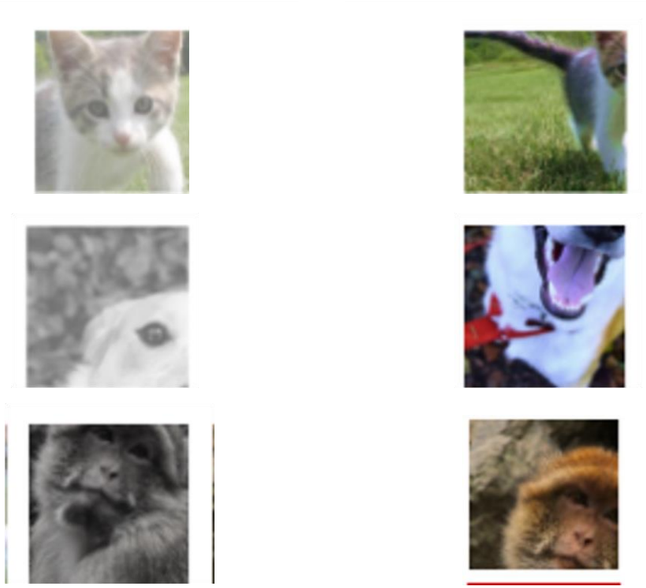




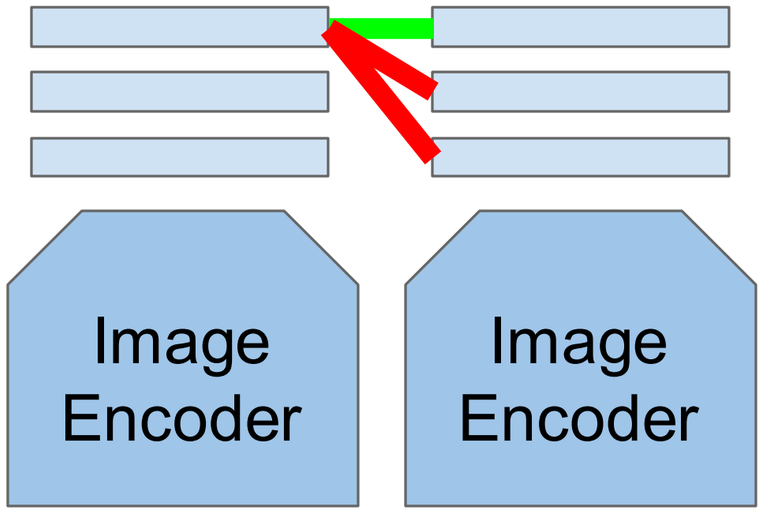


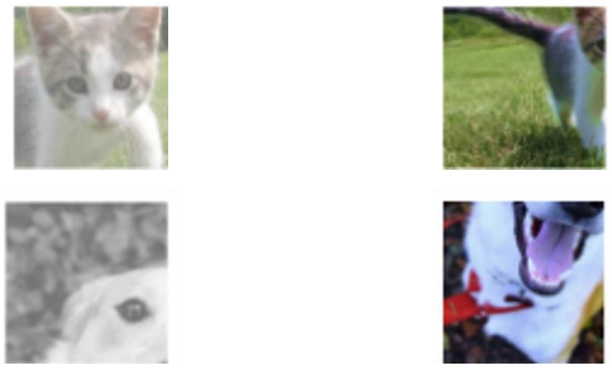
SimClr



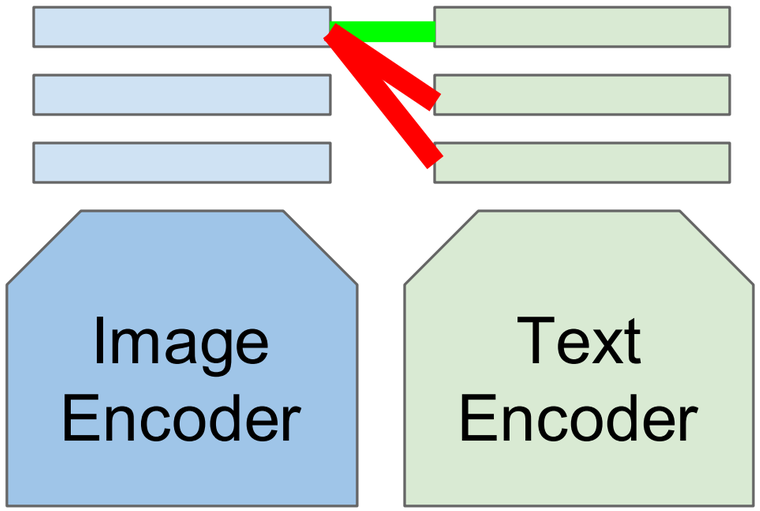


SimClr





CLIP





“My favorite dog is a golden retriever”



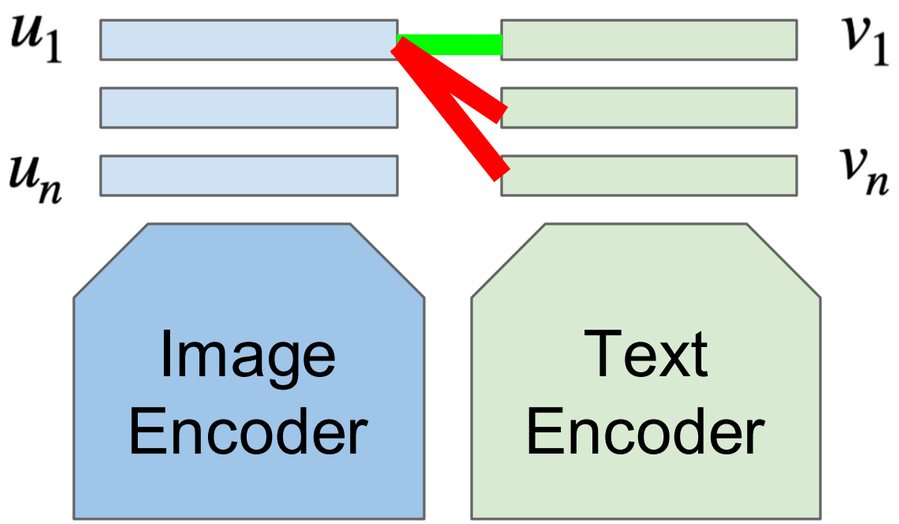
“A cute fluffy cat”



“Monkeys are my favorite animal”

CLIP is trained with the same contrastive objective

$$\sum_{i=1}^{n} -\log\left(\frac{e^{\langle u_i, v_i\rangle}}{\sum_{j=1}^{n} e^{\langle u_i, v_j\rangle}}\right)$$





"My favorite dog is a golden retriever"



"A cute fluffy cat"

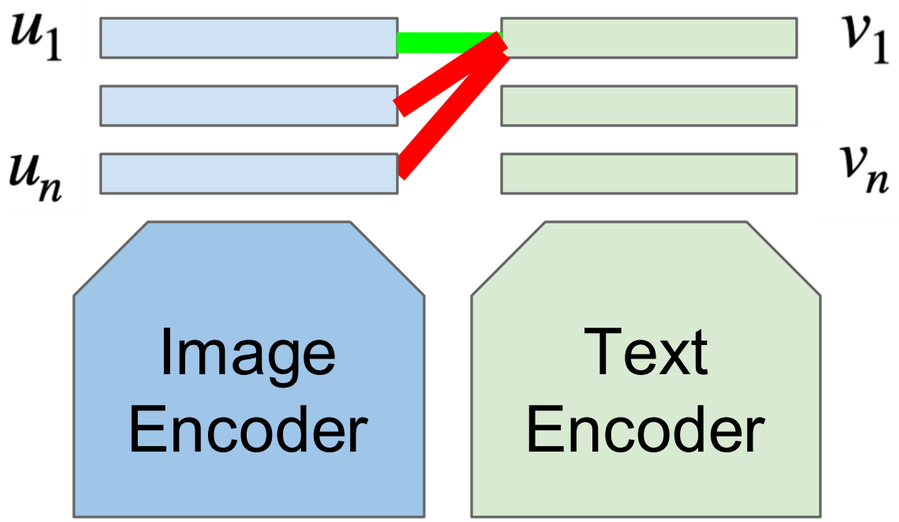


"Monkeys are my favorite animal"

CLIP Training Objective

$$\sum_{i=1}^{n} -\log\left(\frac{e^{\langle u_i, v_i\rangle}}{\sum_{j=1}^{n} e^{\langle u_i, v_j\rangle}}\right)$$

$$+\sum_{i=1}^{n} -\log\left(\frac{e^{\langle u_i, v_i\rangle}}{\sum_{j=1}^{n} e^{\langle u_j, v_i\rangle}}\right)$$





"My favorite dog is a golden retriever"



"A cute fluffy cat"



"Monkeys are my favorite animal"

Lots of image-text data can be found online

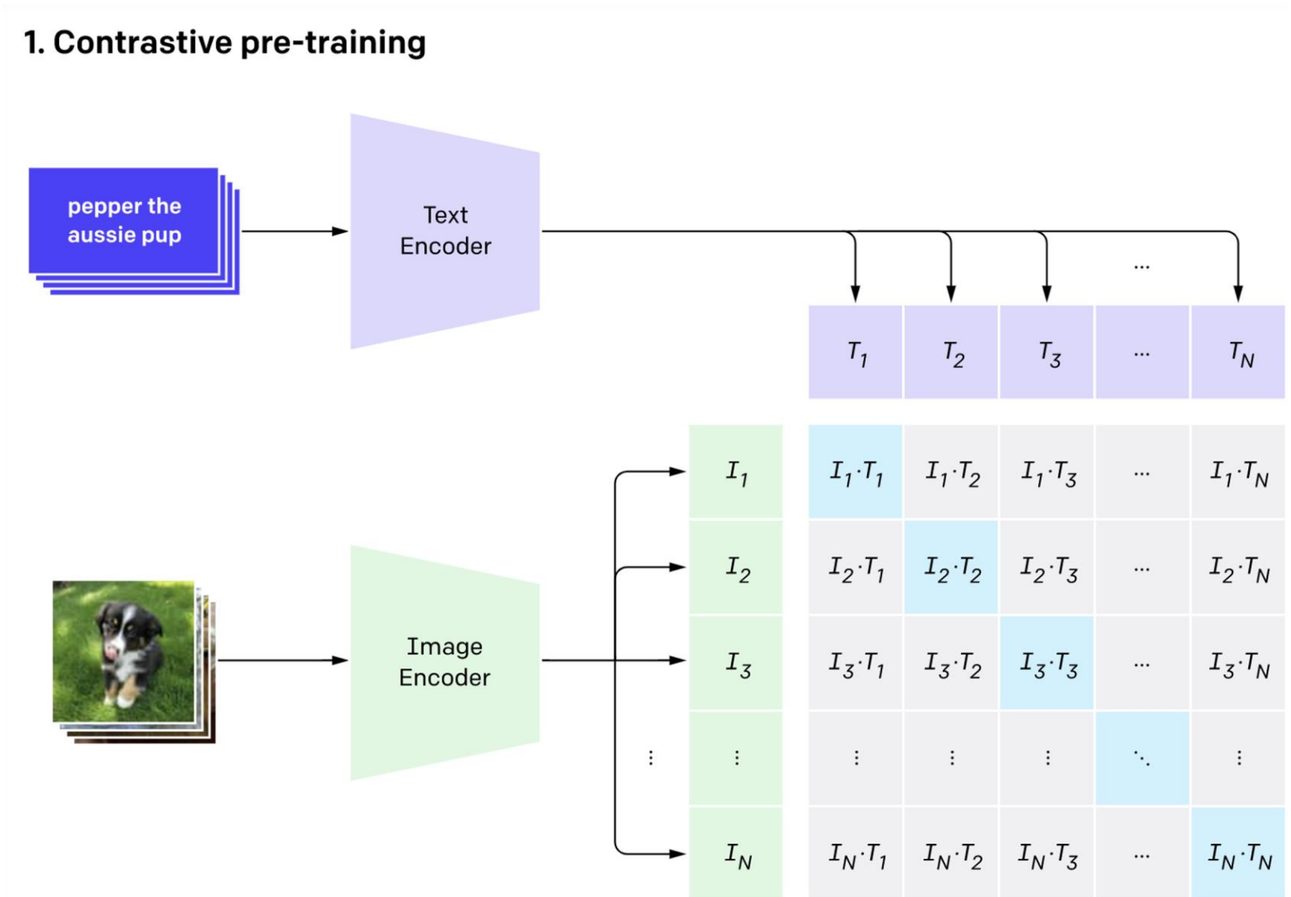


Mount Rainier's northwestern slope viewed aerially just before sunset on September 6, 2020

CLIP training data was scraped at scale from images and their associated alt-text from the internet

https://en.wikipedia.org/wiki/Mount_Rainier

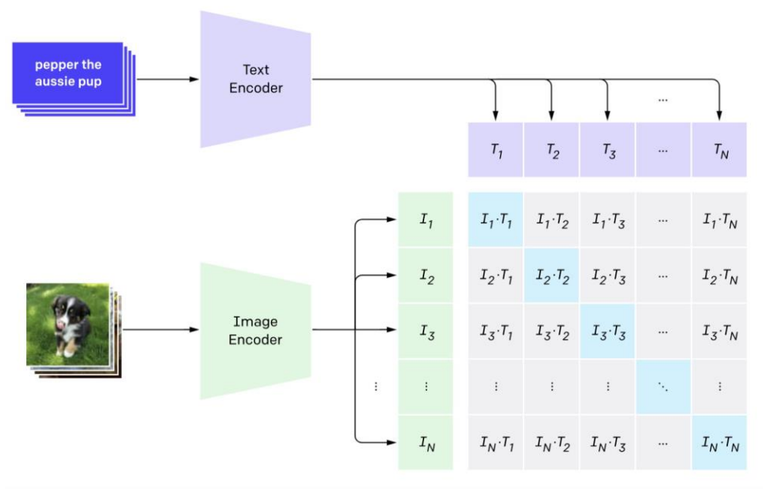
CLIP Training Objective



At the end of training, you have a model that will give you a similarity score between an image and a text

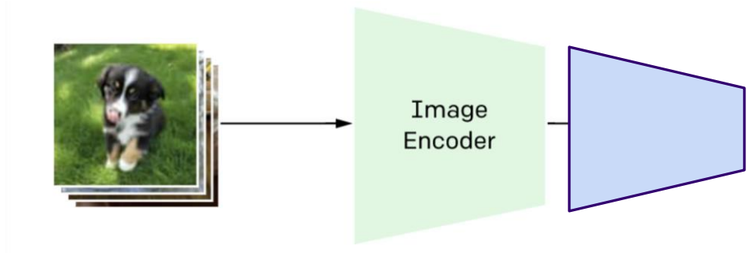
Using pre-trained models out of the box

Step 1: Pretrain a network on a pretext task that doesn't require supervision



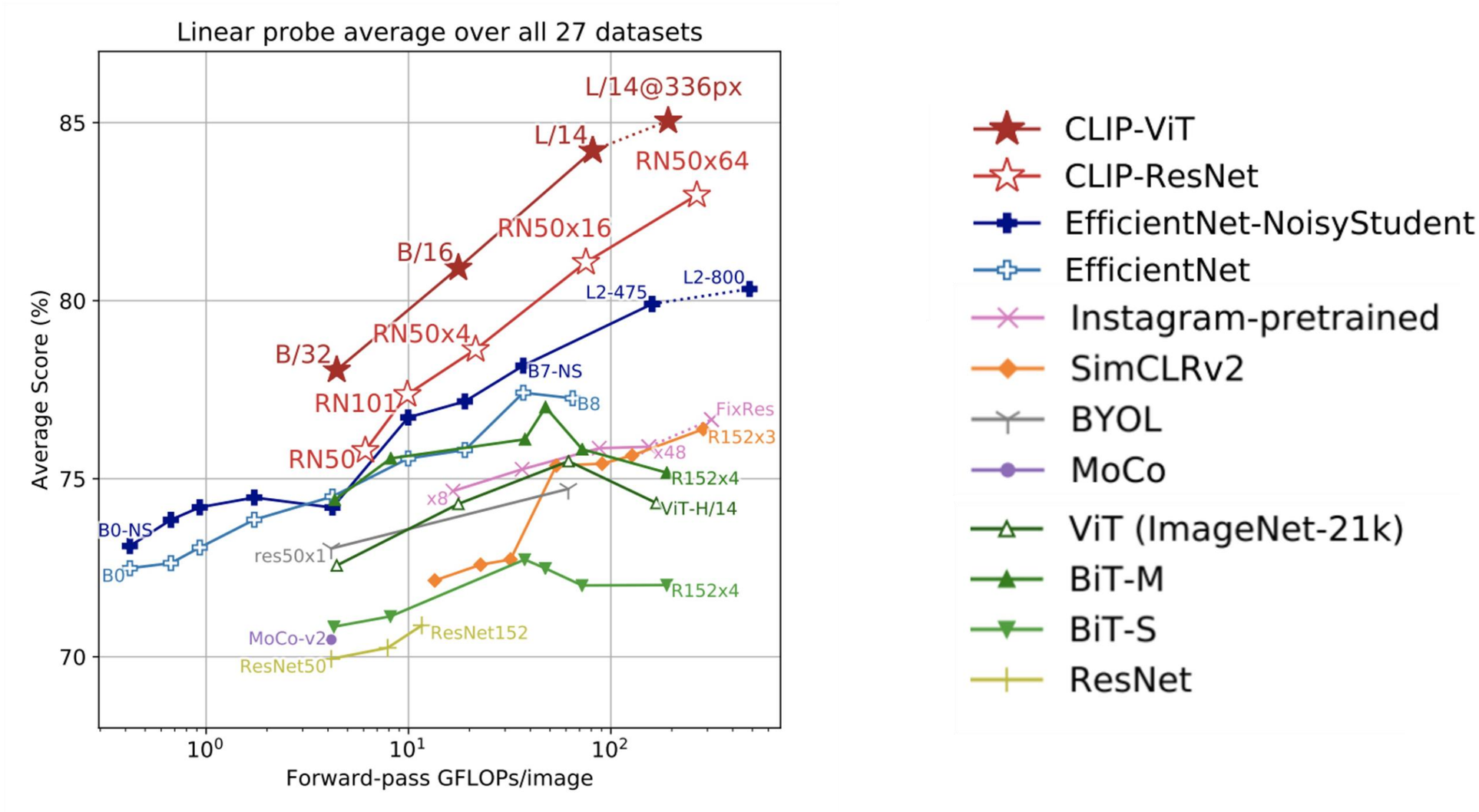
Pre-training tasks:
Contrastive Objective

Step 2: Transfer encoder to downstream tasks via linear classifiers



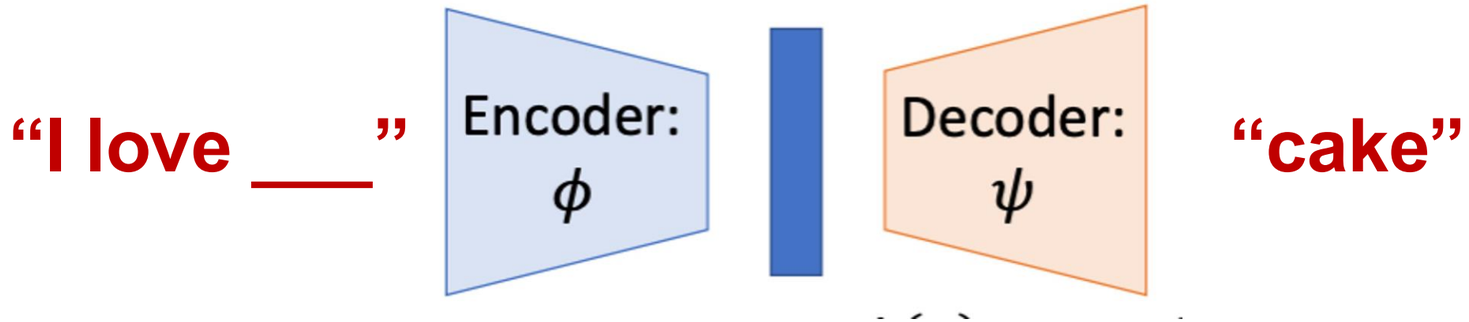
Downstream tasks:
Image classification, object detection, semantic segmentation

CLIP features w/ linear probe across multiple datasets

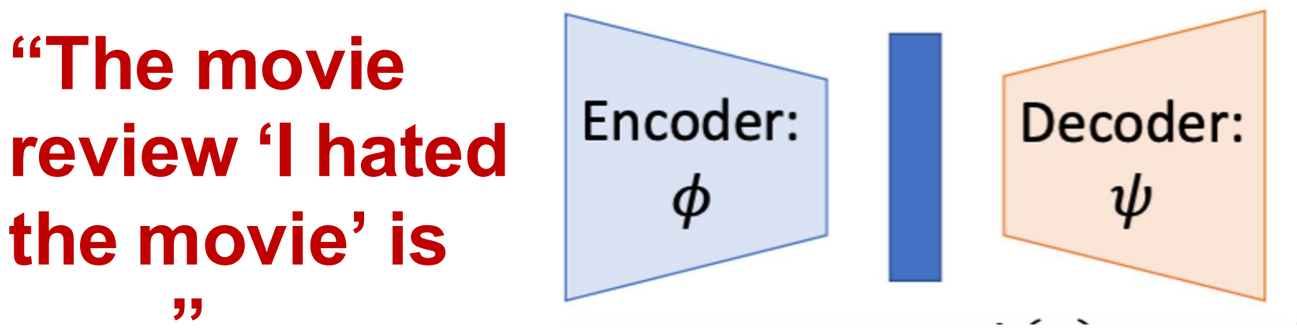


Big difference with language models: We can use LLMs zero-shot for new downstream tasks

Step 1: Pretrain a network on a pretext task that doesn't require supervision

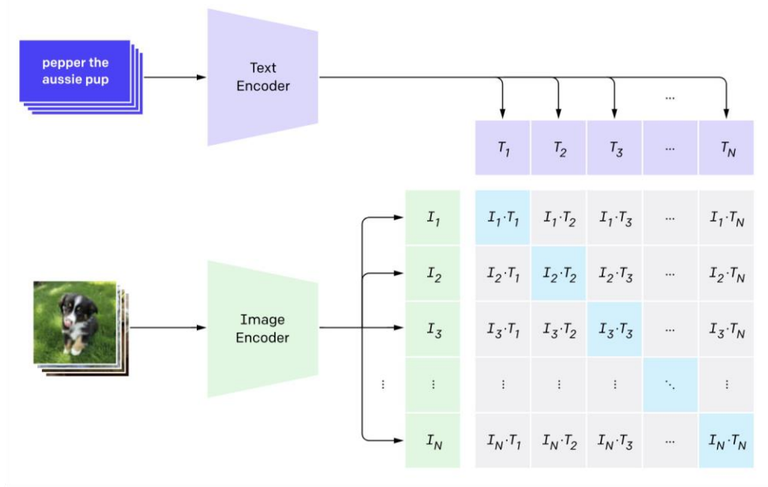


Step 2: Use the model out of the box in a creative way!



But how do we use pre-trained vision-language models in a zero-shot manner?

Step 1: Pretrain a network on a pretext task that doesn't require supervision

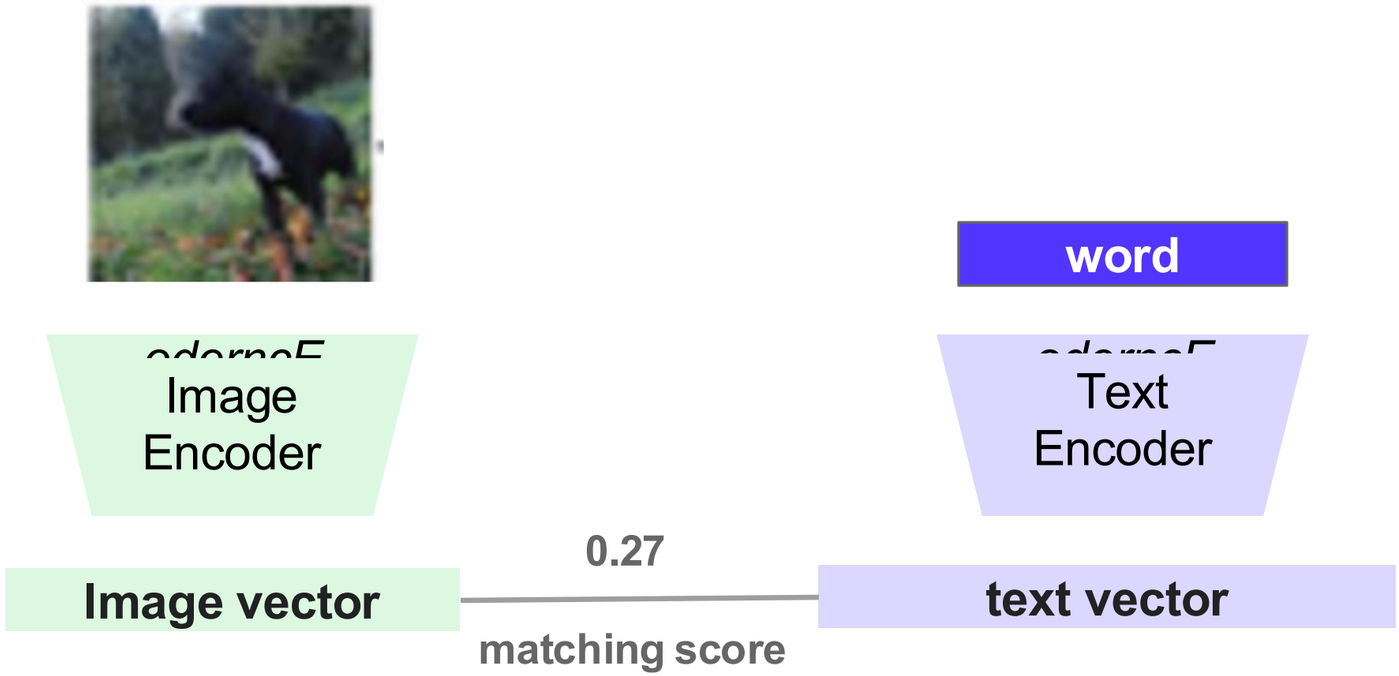


Pre-training tasks:
Contrastive Objective

Step 2: Use the model out of the box in a creative way!

Out of the box classification (No fine-tuning)

Clever trick: we can create a classifier using the text encoder!



Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Create a vector representation for *each* category!



Image Encoder

Image vector

plane

dog

bird

Text Encoder

“Plane” vector

“Dog” vector

“Bird” vector

Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Match a new image to the most similar vector



Image Encoder

plane

dog

bird

Text Encoder

Image vector

0.13

0.27

0.09

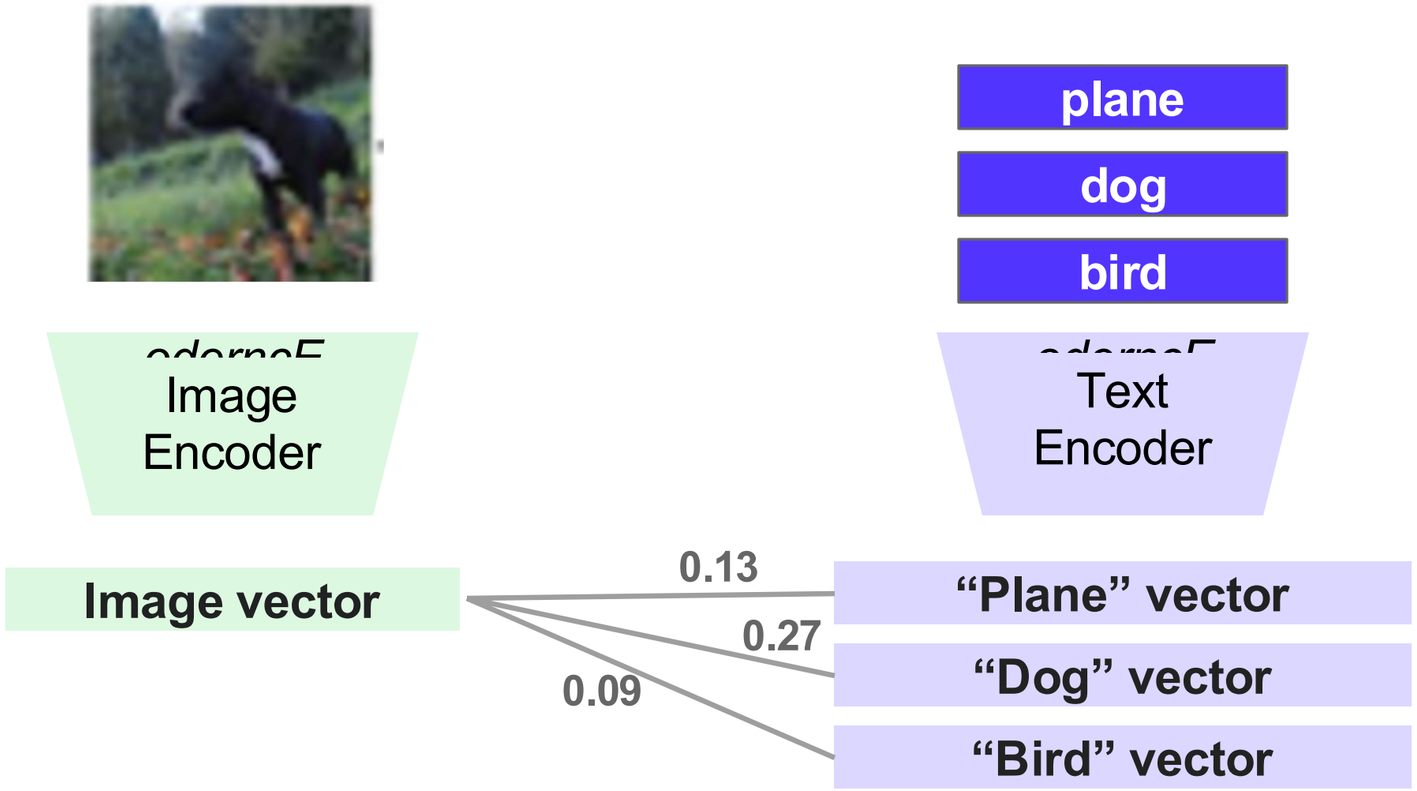
“Plane” vector

“Dog” vector

“Bird” vector

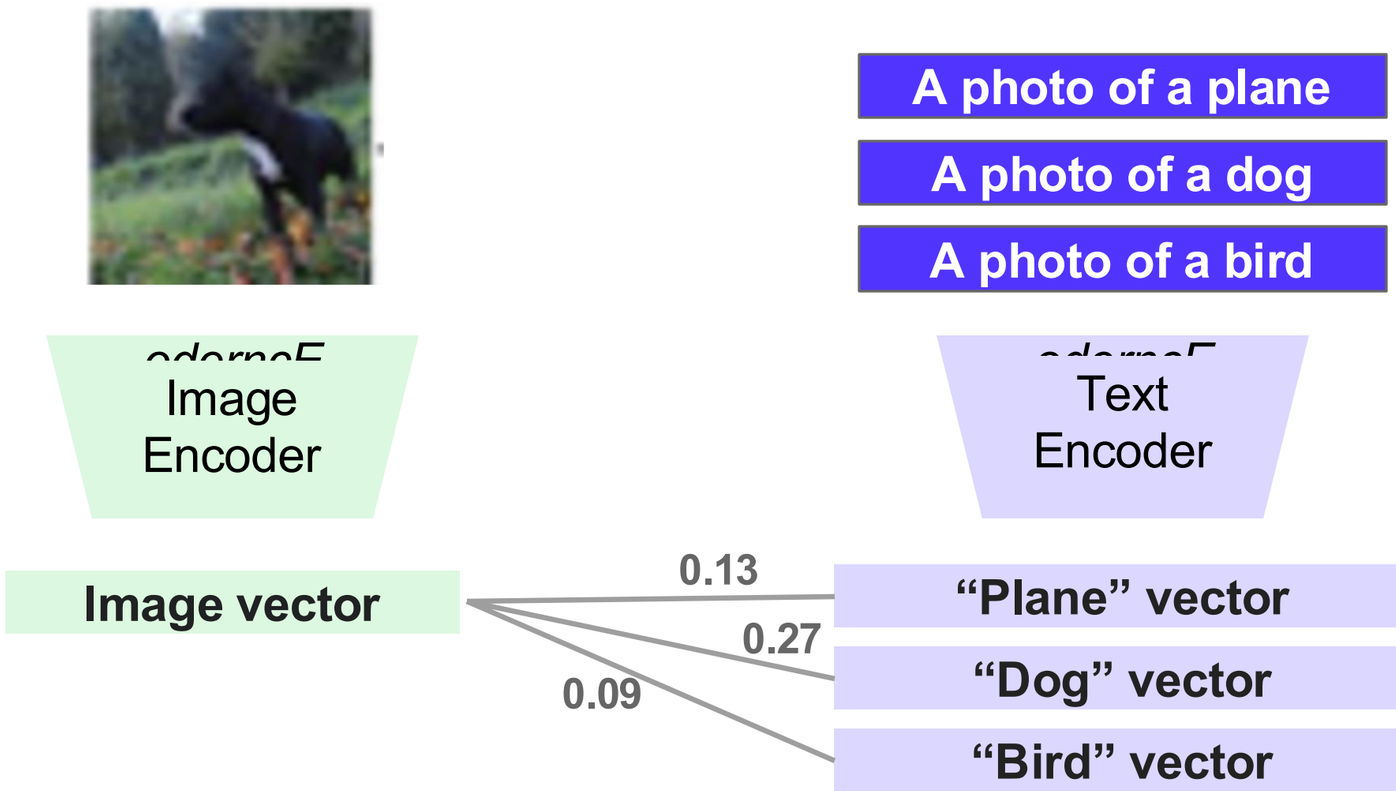
Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

You can think of this as a 1-NN algorithm with the vectors as the training data



Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Since CLIP was trained with phrases, you can improve performance by using a phrase “A photo of a [category]”



+1.3% on ImageNet

Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

A single phrase might be too biased.

Solution: Use multiple phrases



Image Encoder

Image vector

A photo of a plane | A drawing of a plane | …

A photo of a dog | A drawing of a dog | …

A photo of a bird | A drawing of a bird | …

Text Encoder

“Plane” vector 1 | “Plane” vector 2

“Dog” vector 1 | “Dog” vector 2

“Bird” vector 1 | “Bird” vector 2

Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Use the average vector across phrases as the representation for each category



Image Encoder

Image vector

A photo of a plane | A drawing of a plane | …

A photo of a dog | A drawing of a dog | …

A photo of a bird | A drawing of a bird | …

Text Encoder

0.13

0.27

0.09

Mean “Plane” vector

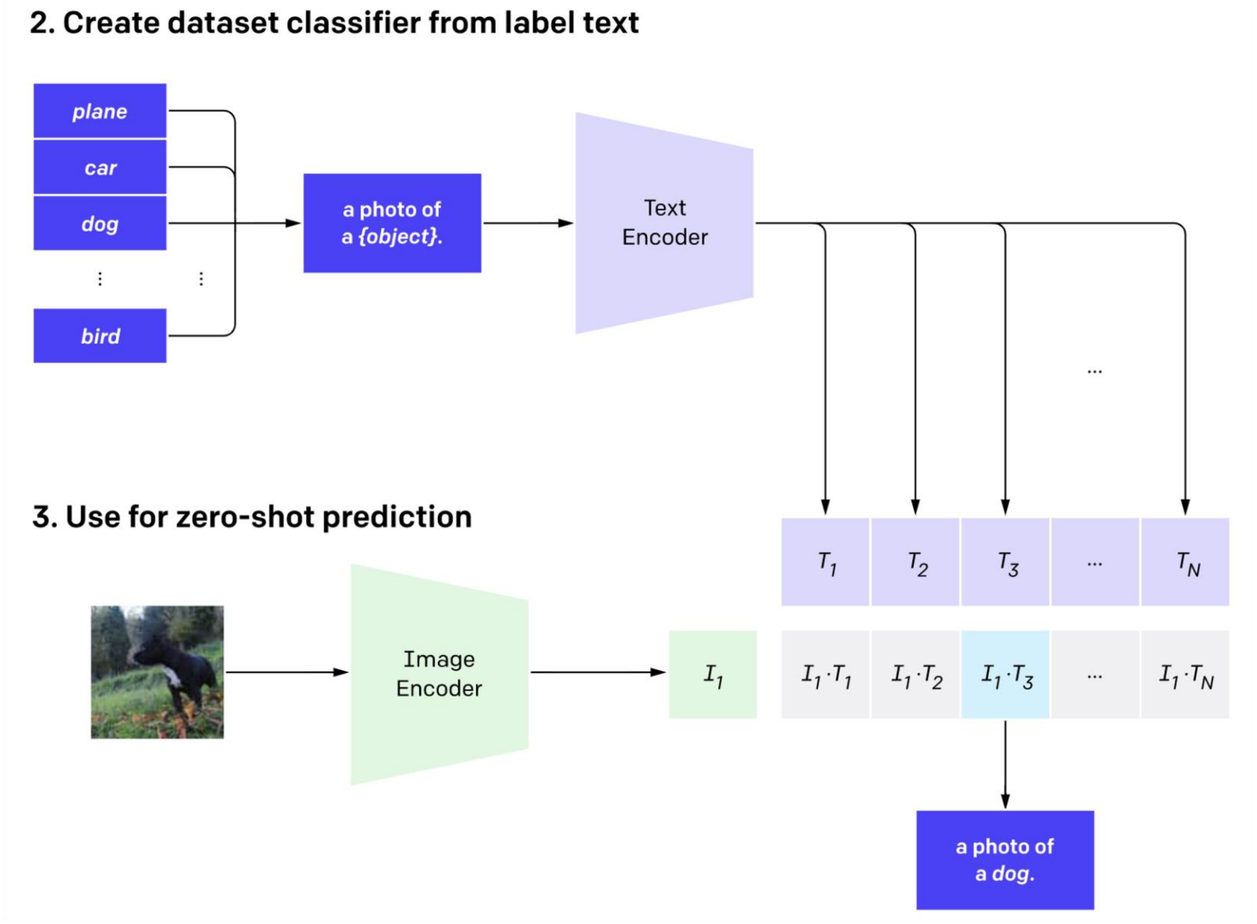
Mean “Dog” vector

Mean “Bird” vector

+5% on ImageNet

Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

That’s it! Now, you can use CLIP as a foundation model for image classification for any dataset



Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Exciting result after training on 400M image-text pairs

DATASET	IMAGENET RESNET101	CLIP VIT-L
	76.2%	76.2%
ImageNet		

Matches the accuracy of of ResNet 101 that has been trained on ImageNet, except CLIP was trained with no human labels at all!

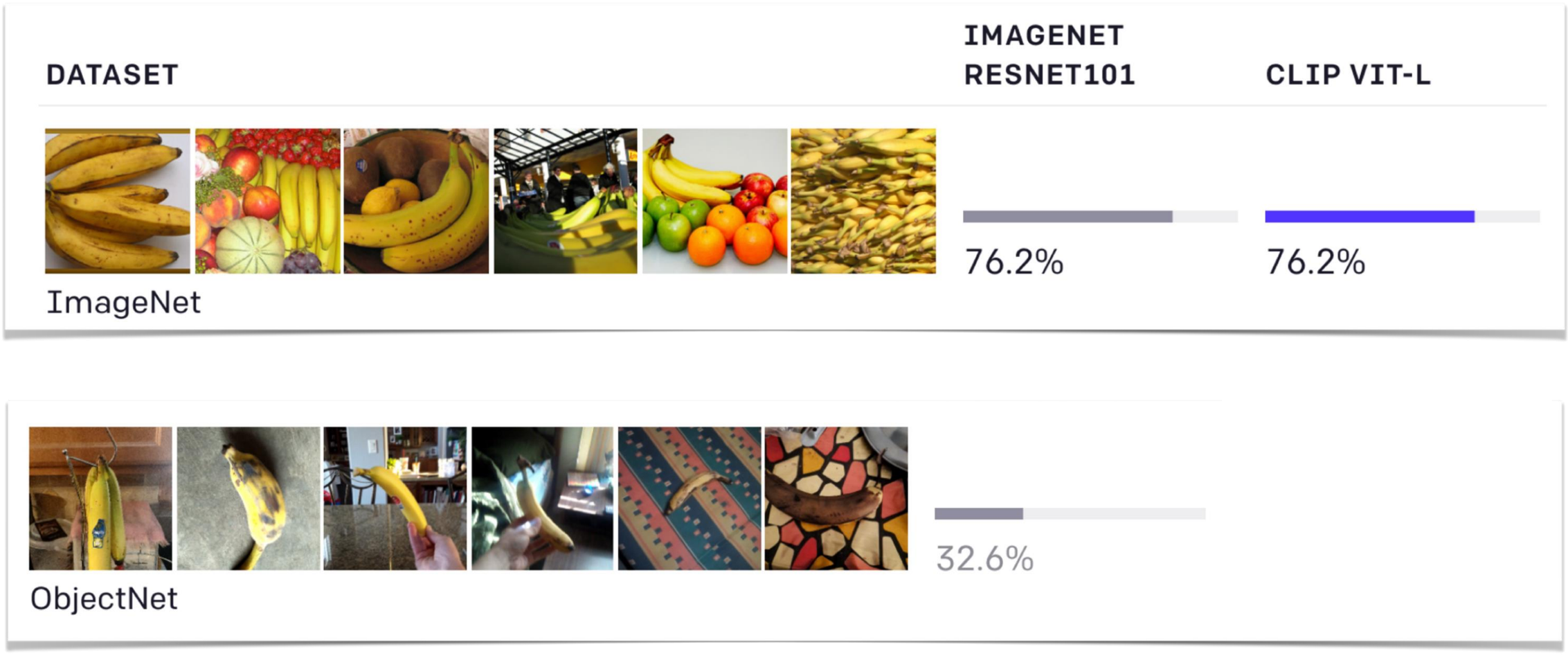
Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Here’s where things get even more exciting

DATASET	IMAGENET RESNET101	CLIP VIT-L
 ImageNet	76.2%	76.2%
 ObjectNet		

Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Training on ImageNet doesn’t generalize to other datasets. ObjectNet contains the same categories but in weird viewpoints



Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

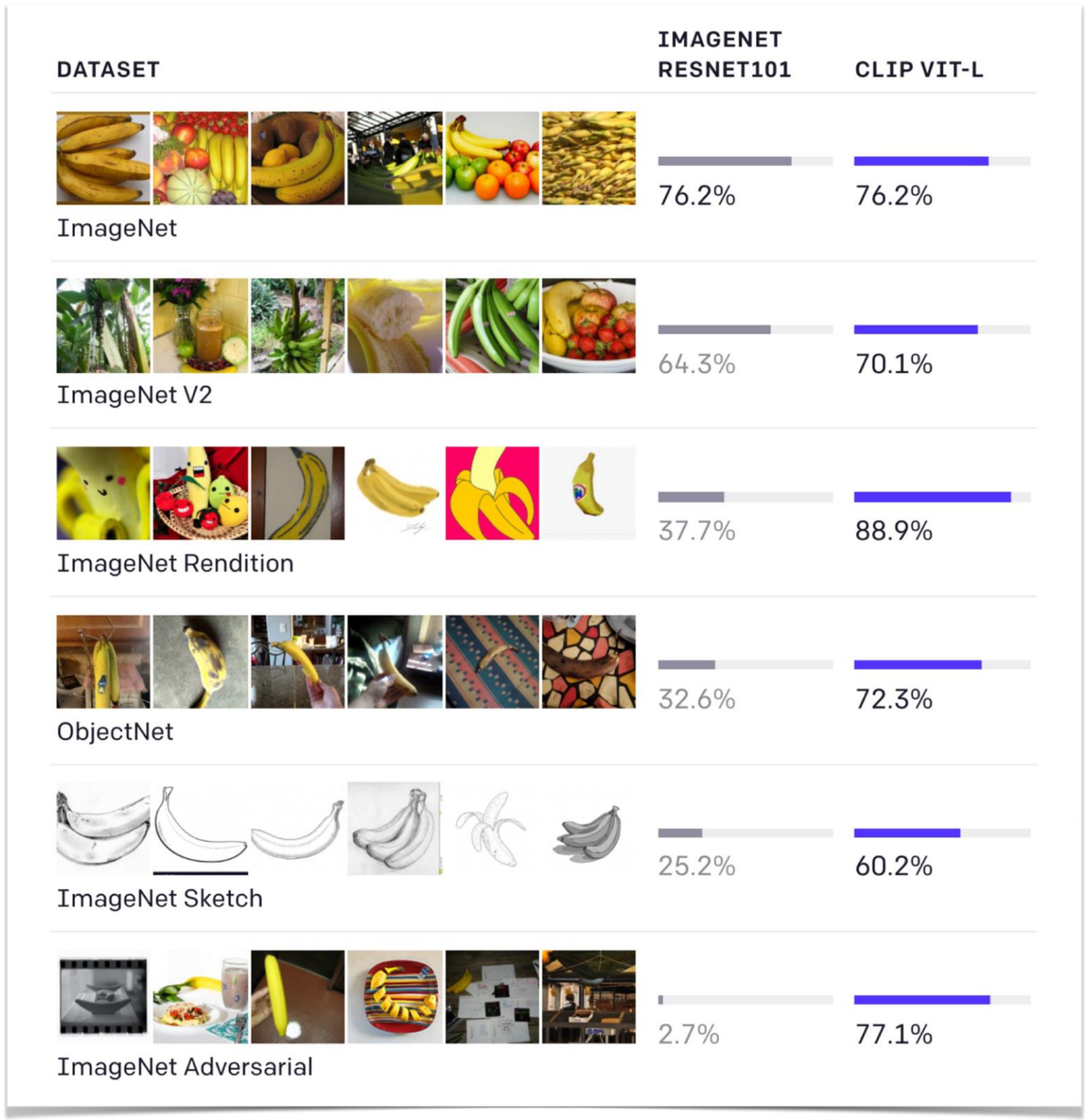
But CLIP zero-shot does so well!

Q. Why do you think that is?

DATASET	IMAGENET RESNET101	CLIP VIT-L
 ImageNet	76.2%	76.2%
 ObjectNet	32.6%	72.3%

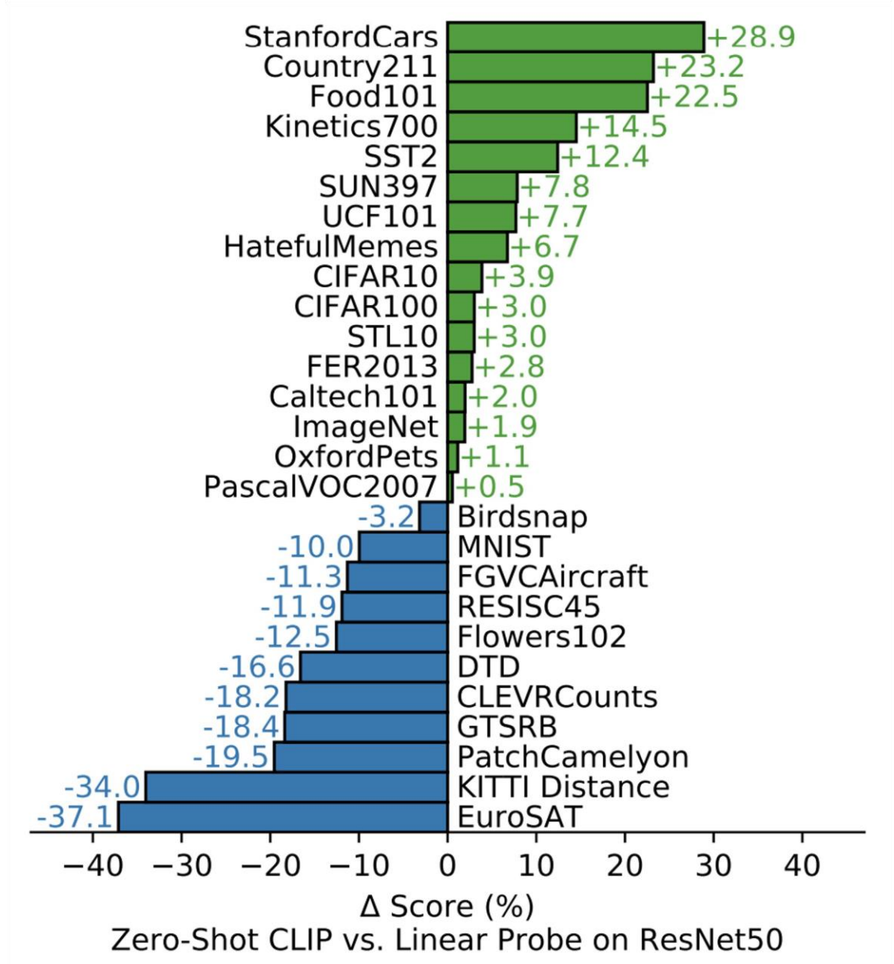
Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

CLIP performance is great also on graphic images , sketches, adversarial datasets,



Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Difference in performance between linear probe vs zero-shot



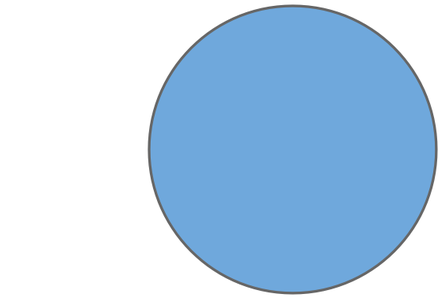
Radford et al “Learning Transferable Visual Models From Natural Language Supervision”

Why does CLIP perform so well?

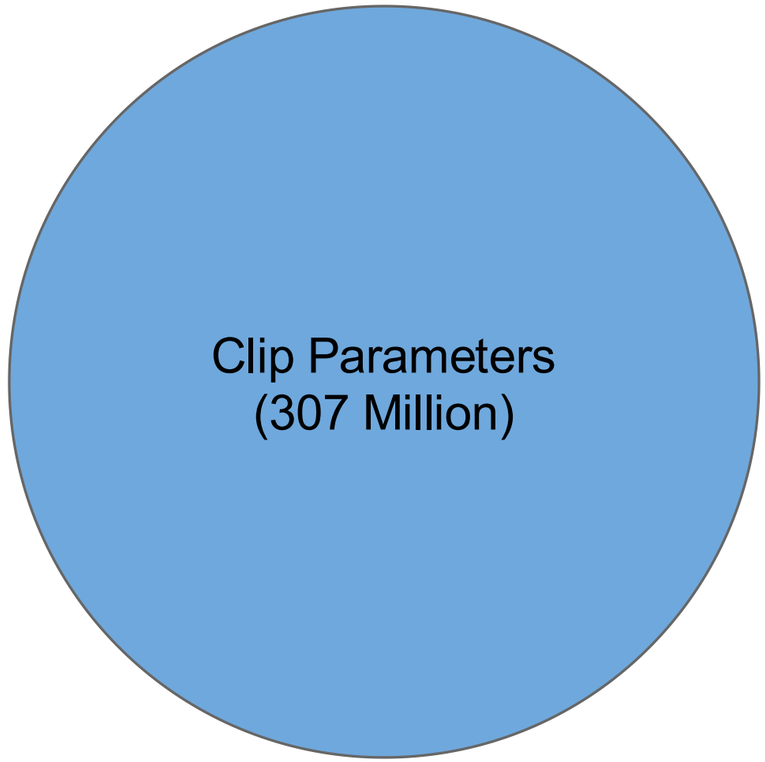
How can no labels beat labels??

Scale!

CLIP scaled up the model parameters with the transformer architecture

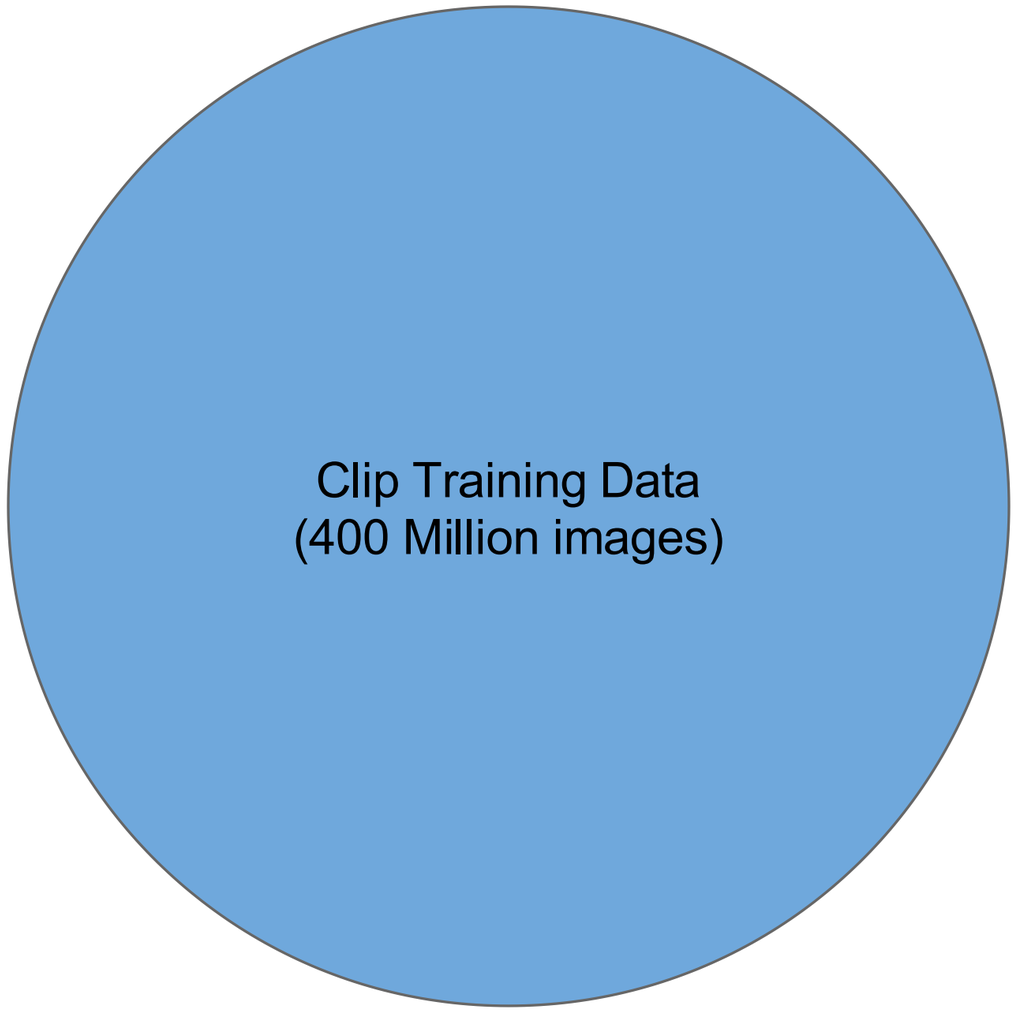


ImageNet ResNet Parameters
(44.5 Million)



CLIP Scaled up the training data by scraping image-text pairs from the internet

ImageNet ResNet Training Data
(1.28 Million)



Advantages of CLIP-style models

1. Dot product is super efficient
    a. Easy to train (enables scaling)
    b. Fast inference, e.g., retrieval over 5B images

2. Open-vocabulary (zero-shot generalization)

3. Can be chained with other models (CuPL)
    [we will discuss this later today]

April 2022, Tristan Thrush et al:

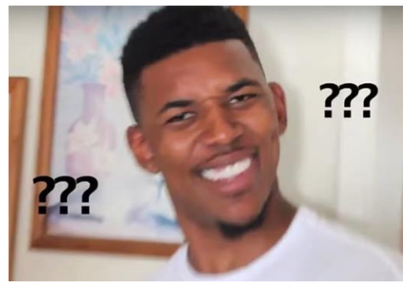
CLIP can’t distinguish between:



there is a mug in some grass



there is some grass in a mug



Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts

Increasing batch size helps you understand fine-grained concepts



Batch size: 4 “animal”

Batch size: 100 “dog”

Batch size: **32000** “Welsh Corgi”

Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts

   Increasing batch size helps you understand fine-grained concepts

   But there's a limit to how fine-grained you can get this way

   Even in a batch of 32K, it's unlikely you see both "a mug in some grass" and "some grass in a mug"

Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts

Winoground



there is a mug in some grass



there is some grass in a mug

“compositionality”

CREPE



Crepe on a skillet. 👁?

Boats on a skillet.

Crepe under a skillet.

Crepe on a dog.

ARO



BLIP

the grass is eating the horse 81%

the horse is eating the grass 78%

…

Paper	Venue	Perturbation
Winoground	CVPR 2022 (Oral)	word order
VL-Checklist	EMNLP 2022	replacements
When-and-Why	ICLR 2023 (Oral)	word order
CREPE	CVPR 2023 (Spotlight)	word order replacements negations
SVLC	CVPR 2023	replacements
DAC	NeurIPS 2023 (spotlight)	replacements
What’s Up	EMNLP 2023	replacements
Text encoders…	EMNLP 2023	word order
SugarCREPE	NeurIPS 2023	word order replacements additions
COLA	NeurIPS 2023 D&B	replacements

Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts

Solution? Hard Negative Fine-Tuning



TODO: Get NegCLIP scores for these captions now

Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts

But training with hard negatives has its own problems…

A black cat and a brown dog
✓

A brown cat and a black dog
✗

A brown dog and a black cat
✗

“hard positives”

Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts
2. Image-level captions are insufficient supervision



“living room”	✓
“house plants”	✗
“couch”	✗

Disadvantages of CLIP-style models

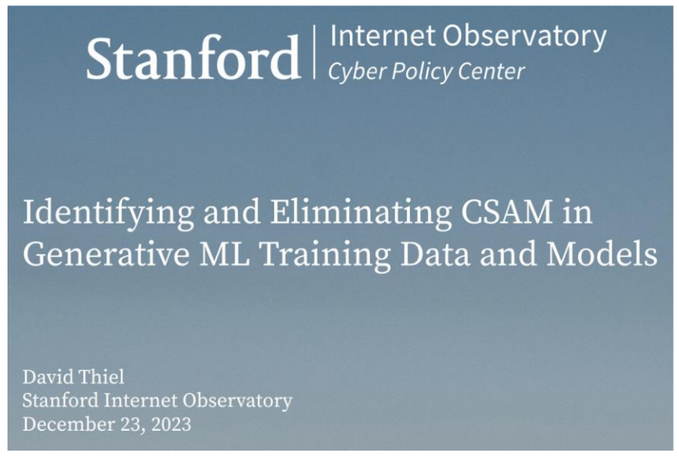
1. Rely too heavily on batch size to learn concepts
2. Image-level captions are insufficient supervision

Also train on region captions with bounding box coordinates



Disadvantages of CLIP-style models

1. Rely too heavily on batch size to learn concepts
2. Image-level captions are insufficient supervision
3. You can't know everything in your 5B dataset



It's extremely important to be intentional about data collection and filtering

Foundation Models

Language	Classification	LM + Vision	And More!	Chaining
ELMo	**CLIP**	**LLaVA**	**Segment Anything**	**LMs + CLIP**
BERT	CoCa	Flamingo	Whisper	**Visual Programming**
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

LLaVA

Motivation: Language models which do next token prediction can be applied to a wide variety of tasks at inference (Math, sentiment analysis, symbolic reasoning)

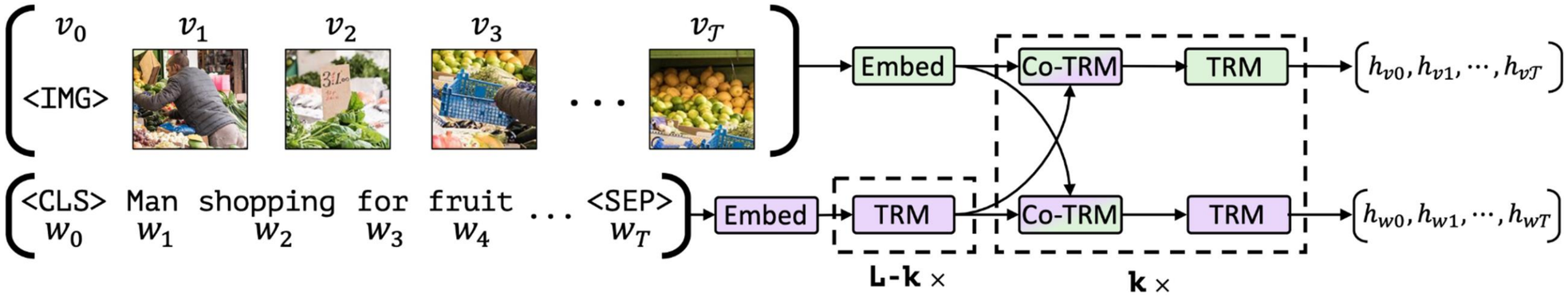
Can we build a model that can accept images and text as input, and then output text?

→ Vision-Language Models

First, some historical context

Vision-Language Models didn’t start with LLaVA!

They go as far back as 2019 → ViLBERT



Historical context

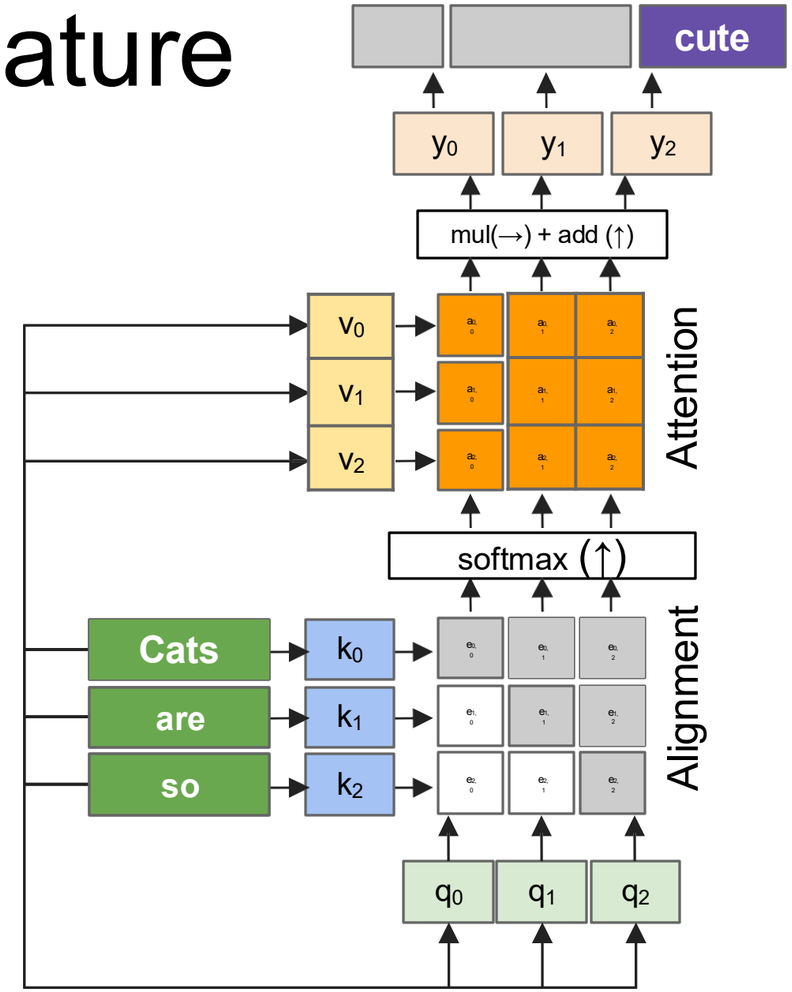
Vision-Language Models didn't start with LLaVA!

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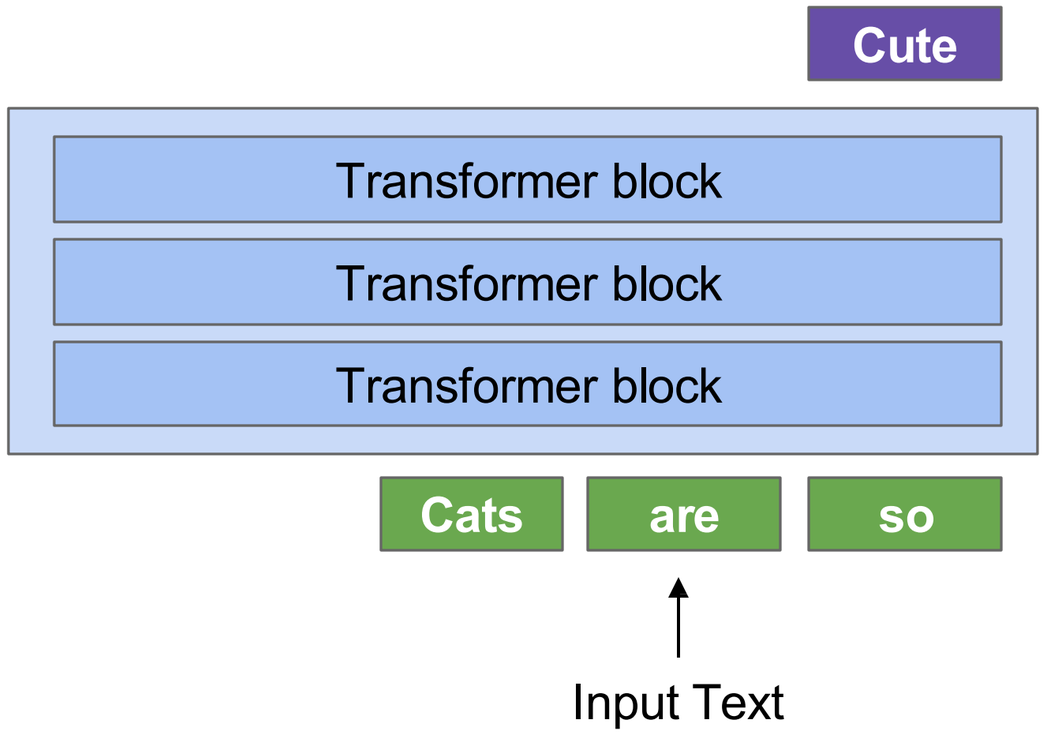
BUT, they had to finetune for each task separately,
with non-trivial task-specific methods (e.g., Mask-RCNN bounding box re-ranking for RefCOCO)

→ Same paradigm as we discussed right at the beginning of this lecture:
very task-specific

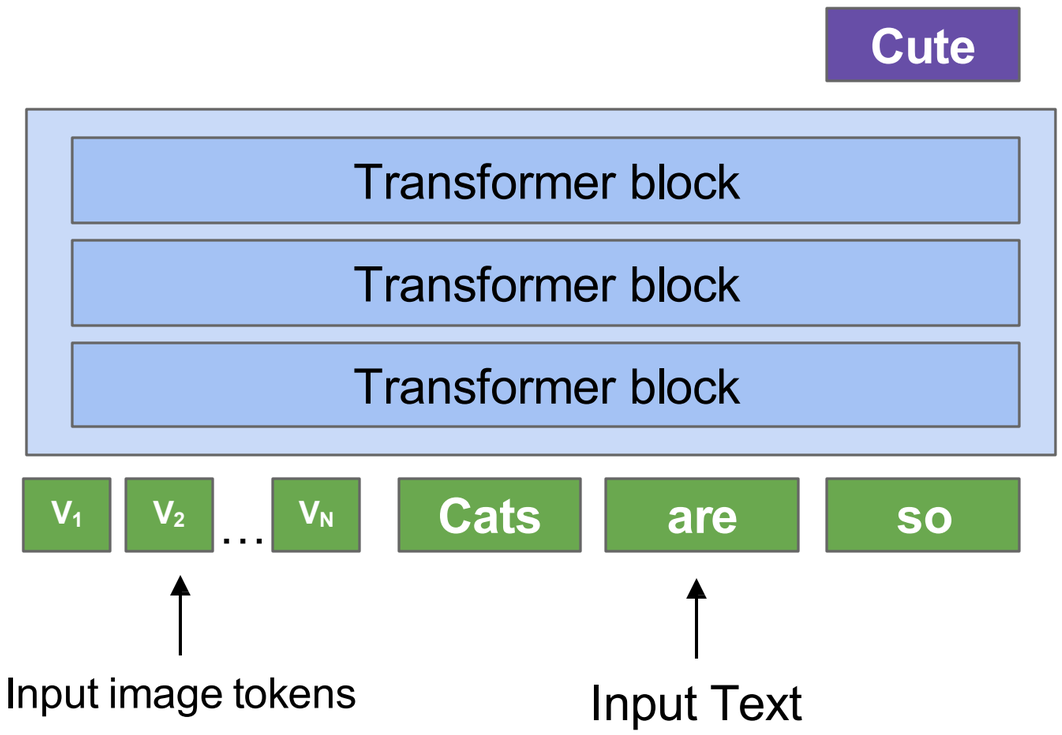
LLaVA uses the autoregressive nature of LLMs



Recall how transformers decode language

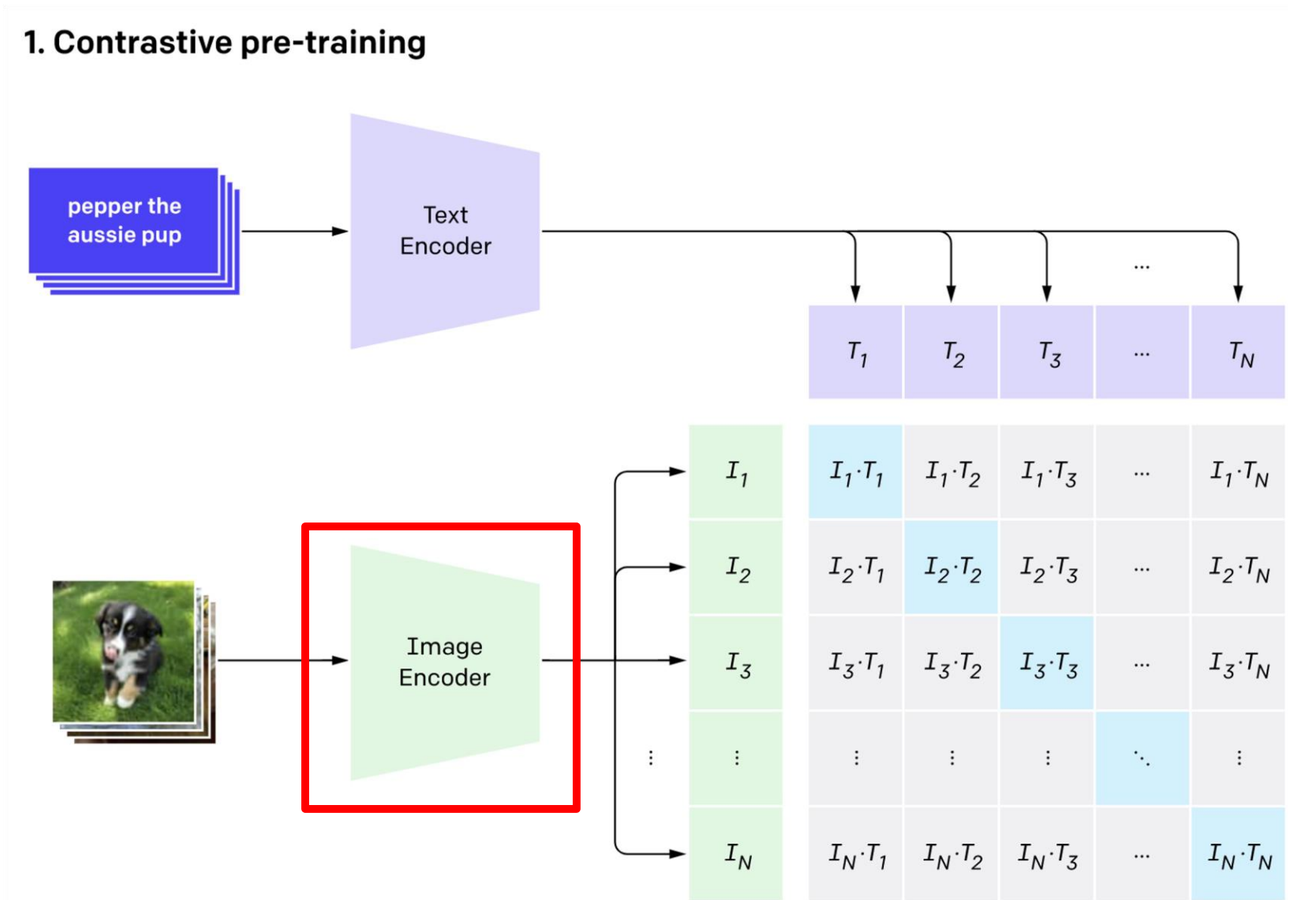


Key idea behind LLaVA – add visual information to the LLM



Which image tokens work best here?

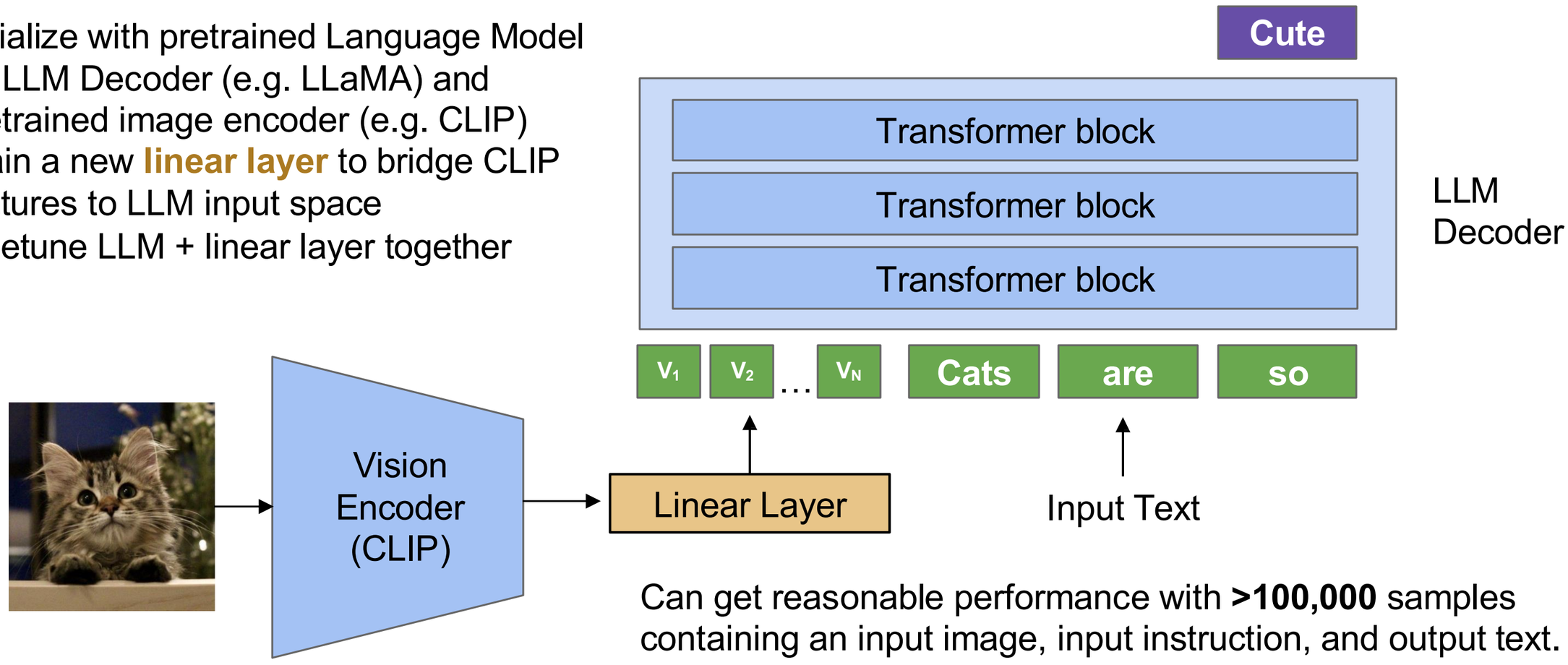
The CLIP encoder is a good option!



At the end of training, you have a model that will give you a similarity score between an image and a text

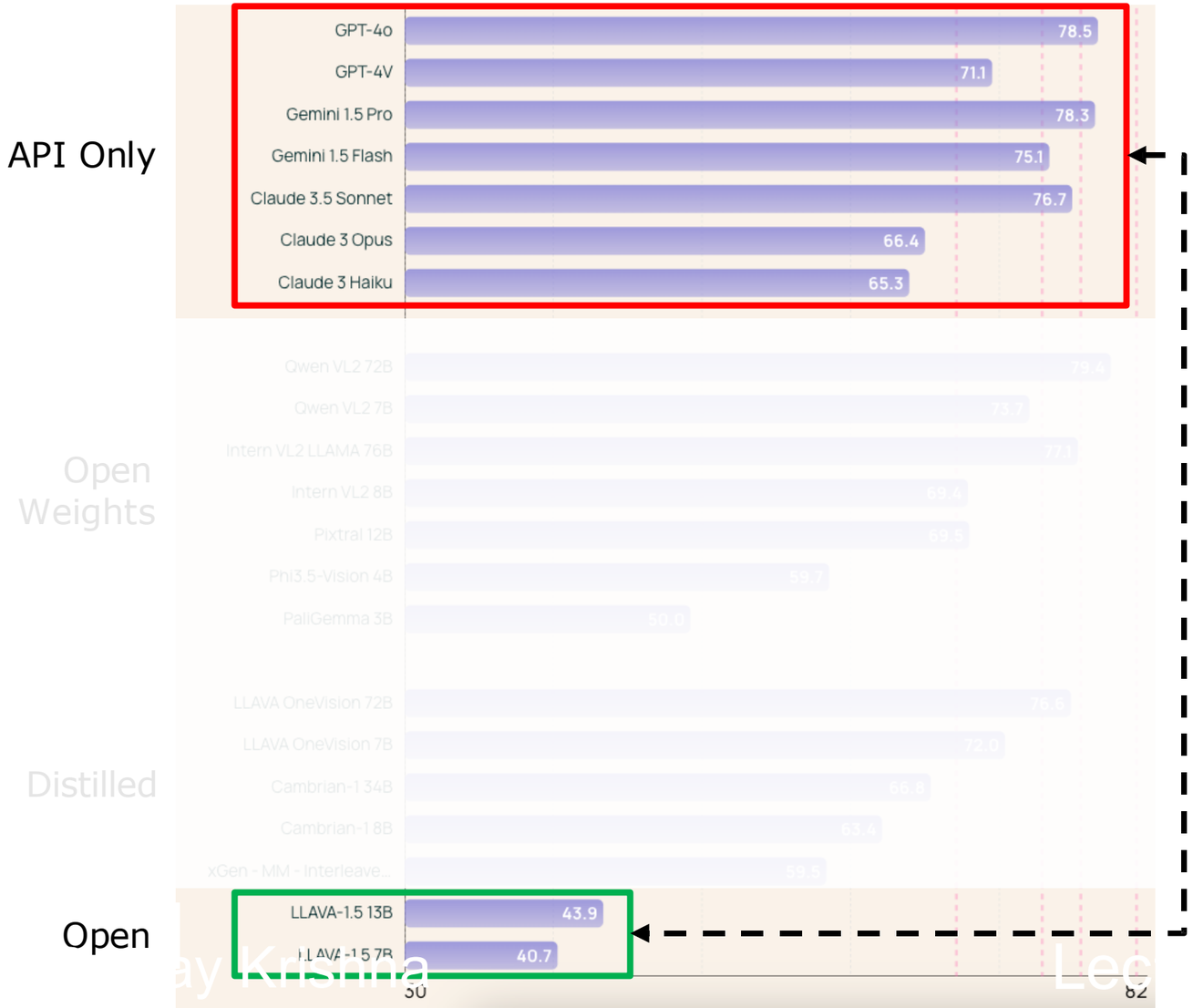
LLaVA – Overall Architecture + Training Recipe

1. Initialize with pretrained Language Model for LLM Decoder (e.g. LLaMA) and pretrained image encoder (e.g. CLIP)
2. Train a new **linear layer** to bridge CLIP features to LLM input space
3. Finetune LLM + linear layer together



Can get reasonable performance with **>100,000** samples containing an input image, input instruction, and output text.

Today, average performance across 11 visual understanding benchmarks





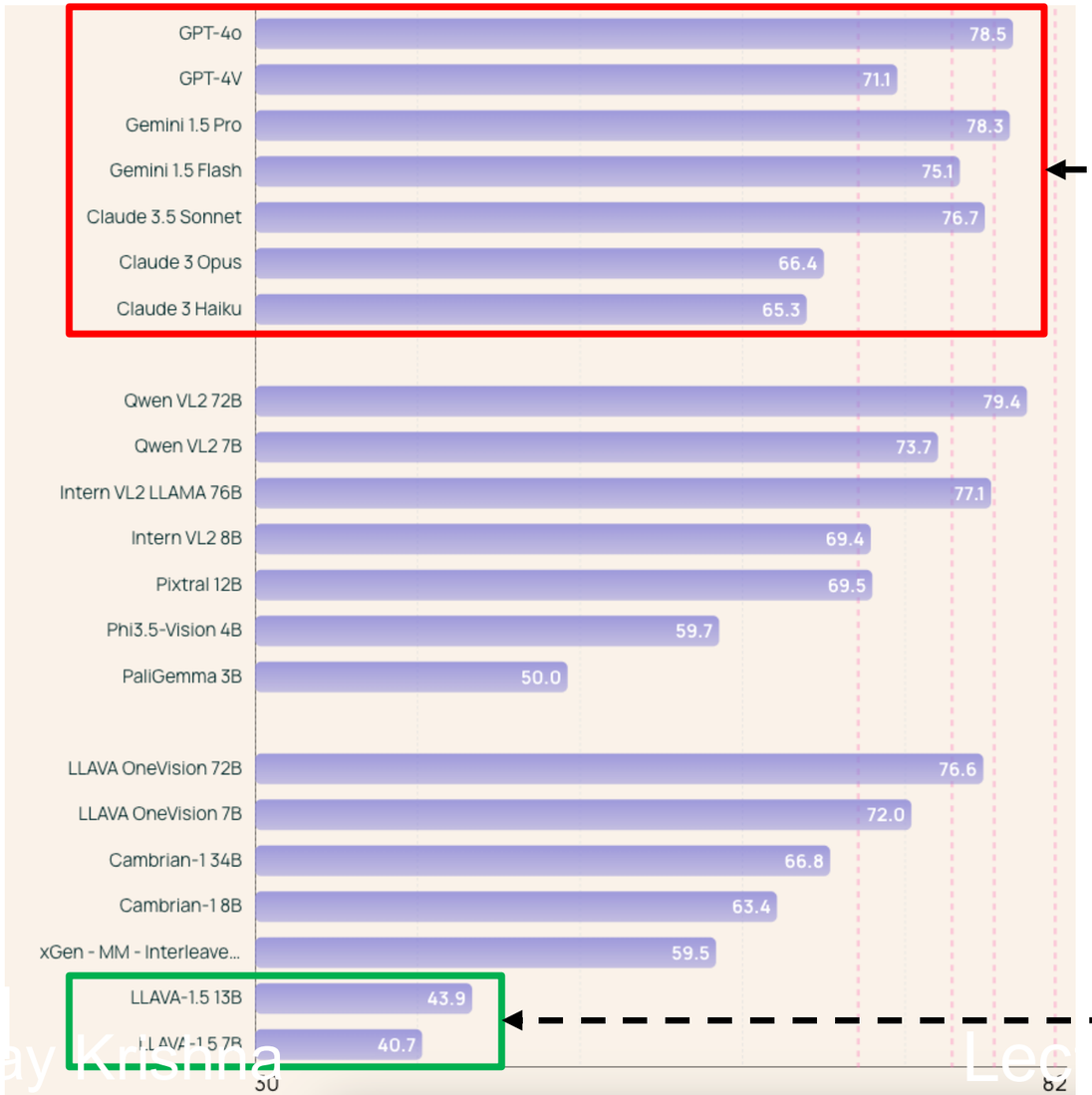
There are open-weight models
but they are all distilled from GPT

API Only

Open Weights

Distilled

Open

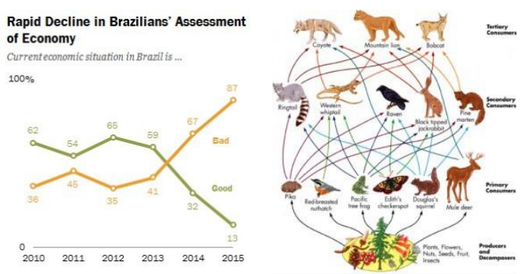














How do we close the gap without relying on proprietary models?

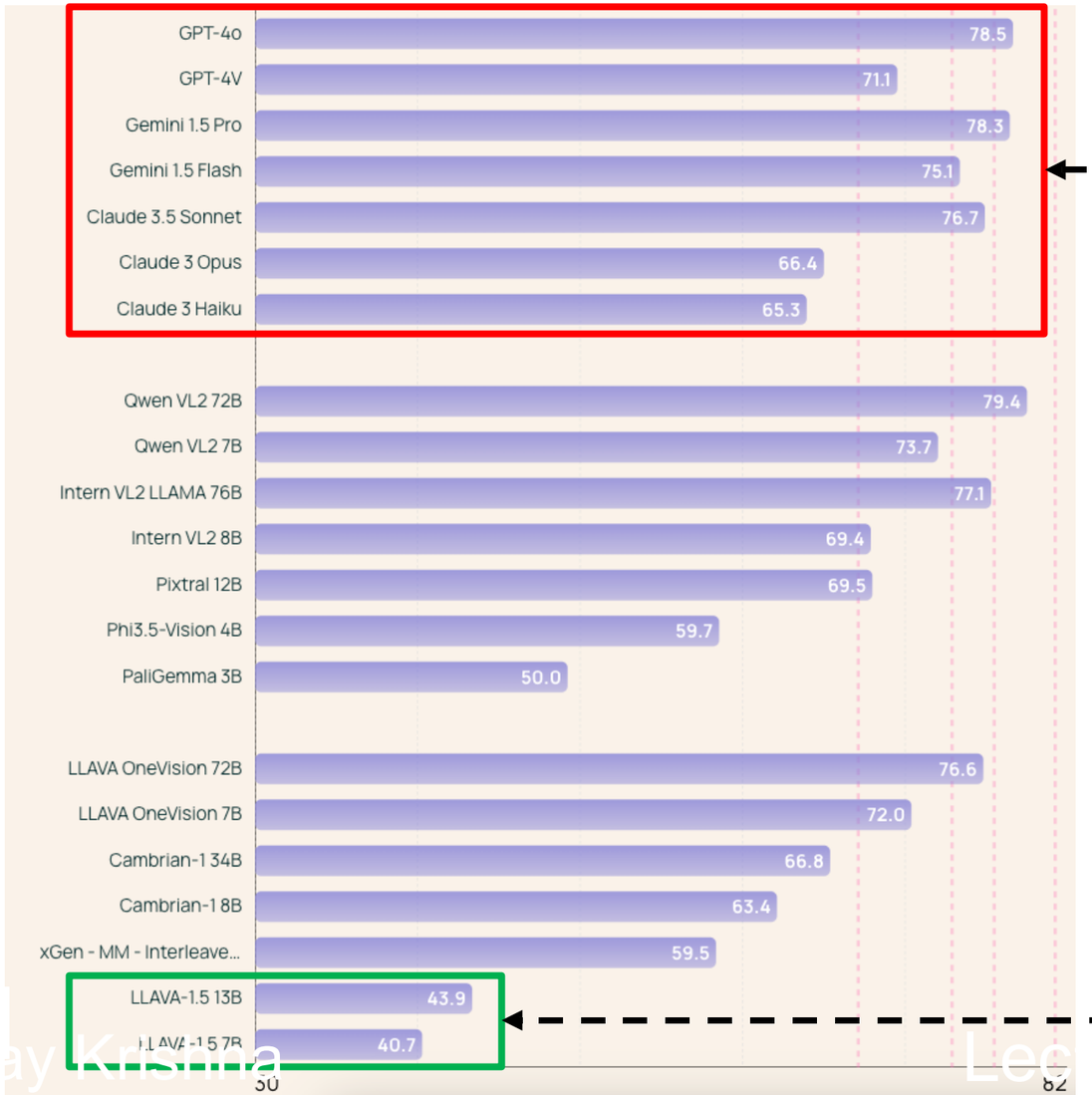
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Open

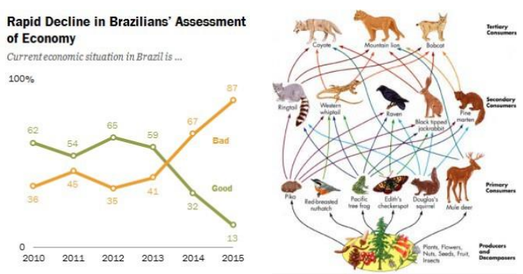














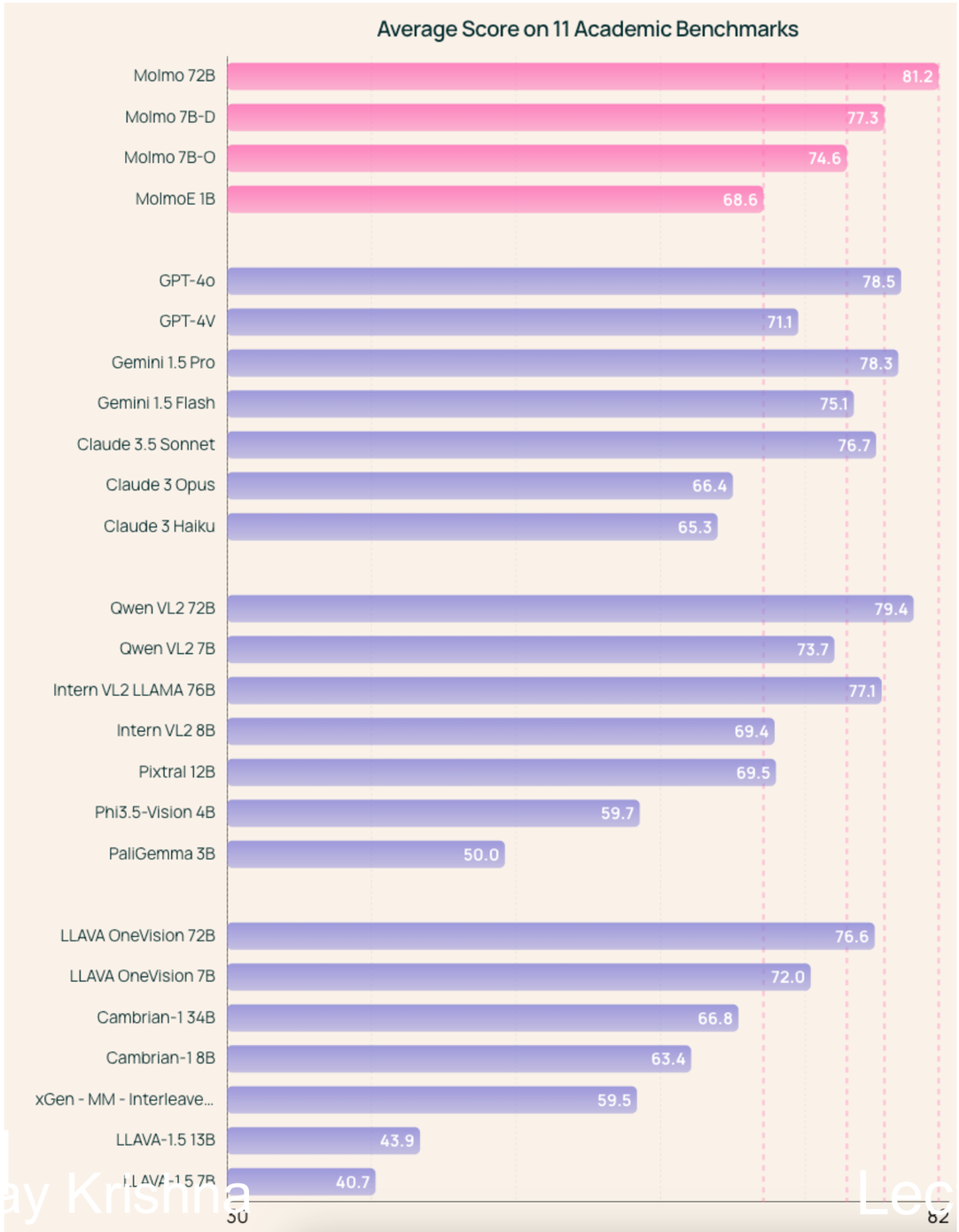
Open
Weights
Data
Code
Evals

API Only

Open Weights

Distilled

Open



Completely
Open
Open Weights
Open Data
Open Code
Open Evals

ay Krishna Lee

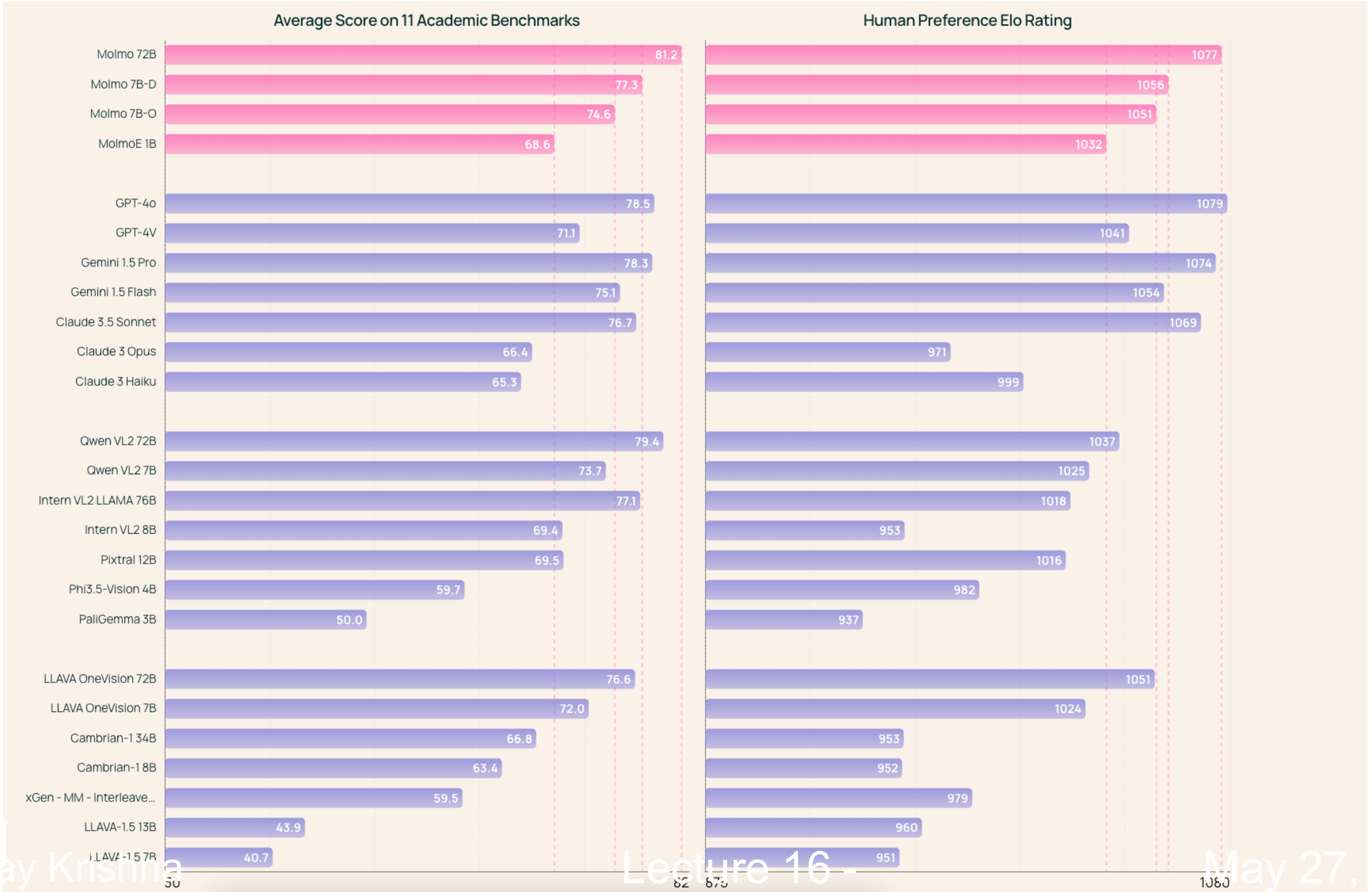
Open
Weights
Data
Code
Evals

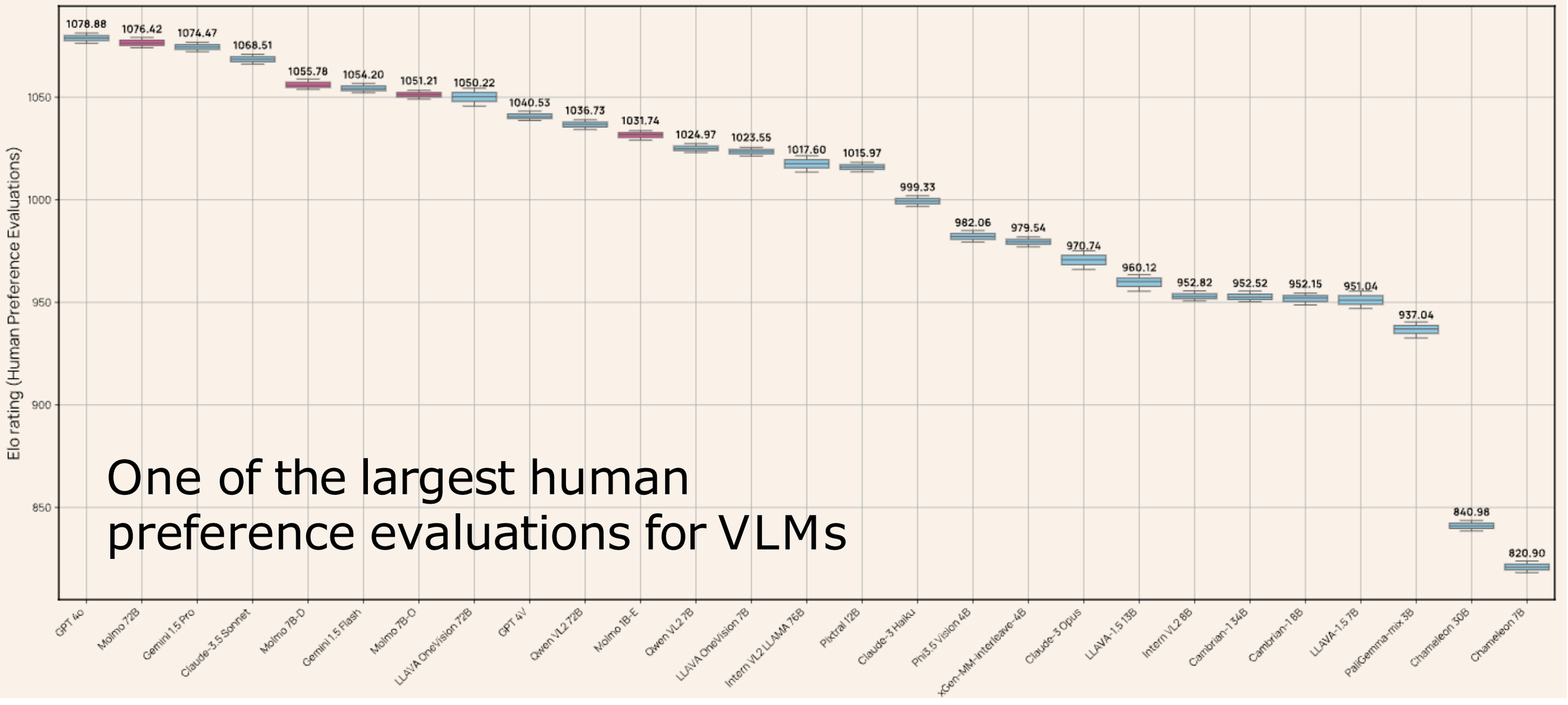
API Only

Open Weights

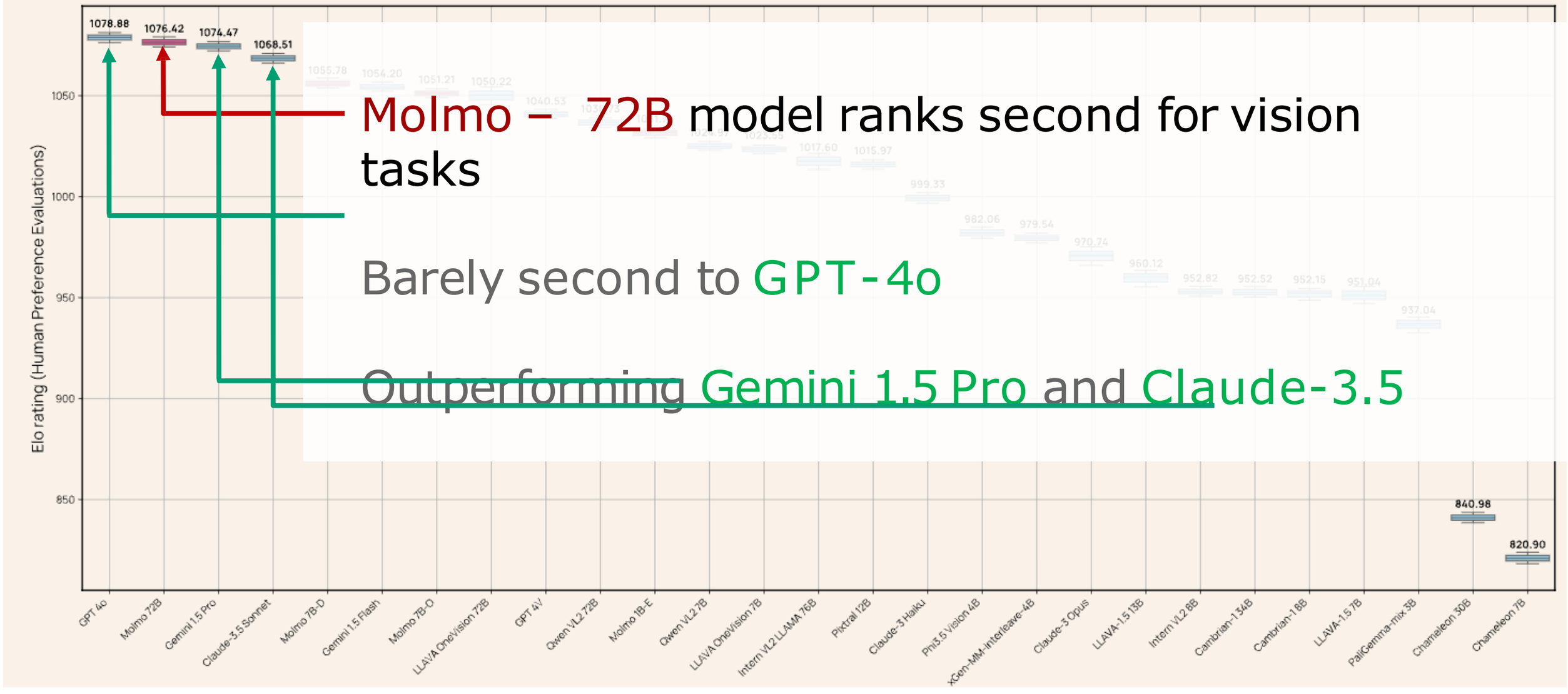
Distilled

Open

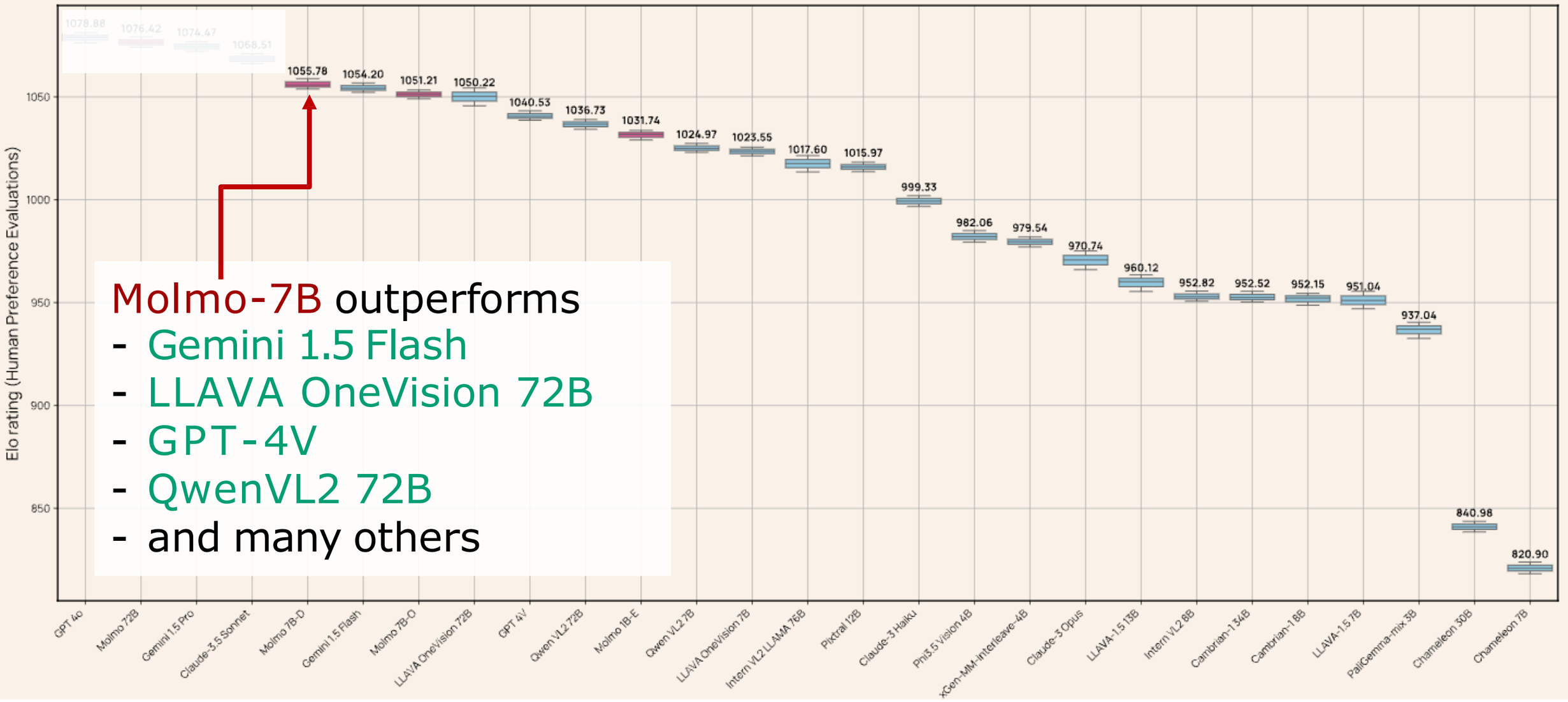




with 325k pairwise comparisons
and 870 human annotators



with 325k pairwise comparisons
and 870 human annotators



Molmo-7B outperforms
- Gemini 1.5 Flash
- LLAVA OneVision 72B
- GPT-4V
- QwenVL2 72B
- and many others

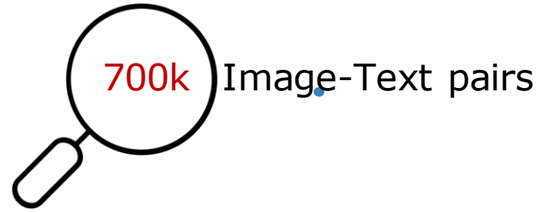
with 325k pairwise comparisons
and 870 human annotators

Data matters! Quality over quantity even for pretraining



Molmo is trained with

PixMo



Internet data is incidental

Human annotated data is intentional



pink, japan,
aesthetic image



love this winter picture by
person

PixMo data is intentional:

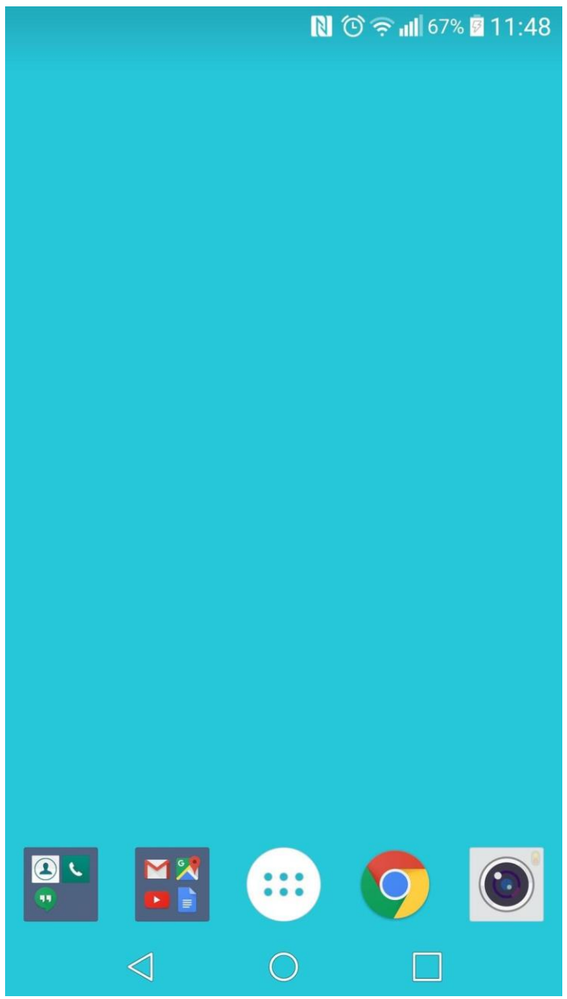


This photograph captures a well-organized work desk set prominently in the middle of the frame. The desk is large and rectangular, made from a polished, rich wood that spans horizontally across the image. Its structure is supported by four distinctive A-shaped legs, adding an elegant touch. On the desk, a striking dual-monitor setup is noticeable: a tall, vertical screen placed behind and to the right of a wider, horizontal computer monitor.

To the right of these monitors, a black mouse rests on a mouse pad. Scattered around the mouse pad, some white papers or letters are strewn across the far right side. On the left side of the desk, a black desk lamp with an extended arm hangs down, illuminating the workspace. Nearby, a stack of books is neatly placed in the upper left corner of the table.

The background wall is painted a subtle beige-white, complementing the refined ambiance of the space. The floor below the desk features elegant pinkish marble tiles, enhancing the room's sophisticated look. To the far right of the image, a large window or patio door allows natural light to pour in, with clear glass that offers a glimpse into the outside area. This exterior view includes part of a rustic brick wall and a metal pail, hinting at an adjacent patio.

Collecting dense captions is hard!!!



This image features a screenshot taken from a tablet device. At the top-right corner, the time is displayed as 11:48, alongside a battery icon indicating 67% power remaining. The device also shows a telephony signal strength of four out of five bars and roughly three-quarters Wi-Fi connectivity. Additionally, a clock alarm icon is present, as well as an icon resembling a white rectangle with a blue letter "N" in it, whose specific function is unclear.

The main portion of the screen, approximately 80% of it, is a solid medium blue color devoid of any content. At the bottom of the screen, there are several folders and icons representing various apps and functionalities:

1. The first folder contains three icons:
- An icon likely for contacts.
- An icon probably for telephone.
- An icon that seems to represent a text messaging app.

2. The second folder houses four apps:
- A Gmail app icon indicated by a red "M" on a white background.
- A YouTube app icon characterized by a red play button.
- A map app icon depicting a map.
- An unidentified app icon represented by a blue folder with its top corner bent down on the right.

3. The third icon is a white oval with six dots, arranged in two rows of three, likely representing an app drawer or menu.

4. The fourth icon resembles a red, yellow, and green shutter with a blue dot in the middle, suggesting it might be for a camera or photo viewing app.

5. Next to it, there is a silver colored camera icon with a black lens and a blue spot in the center, hinting at a camera application.

At the very bottom of the screen, there are three navigational icons:
- A left arrow triangle in white at the bottom-left corner.
- A white oval in the center, indicative of a home button.
- A white rectangle on the bottom-right corner, likely for accessing recent apps or multitasking.

Overall, this image captures the home screen of a tablet, providing an overview of available functionalities and connectivity status.

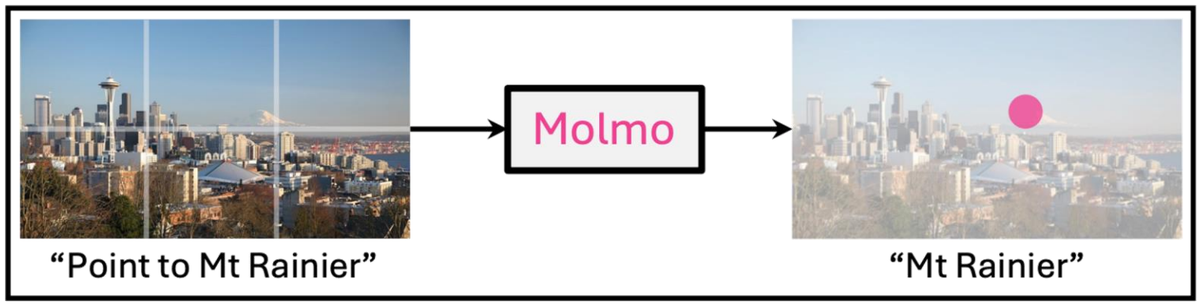
Questions designed to extract meaningful visual information from annotators

- What is the image at first glance?
- What are the objects and their counts?
- What does the text say?
- What are the positions of the objects?
- What subtle details are noticeable?
- What is in the background?
- What is the style and color?

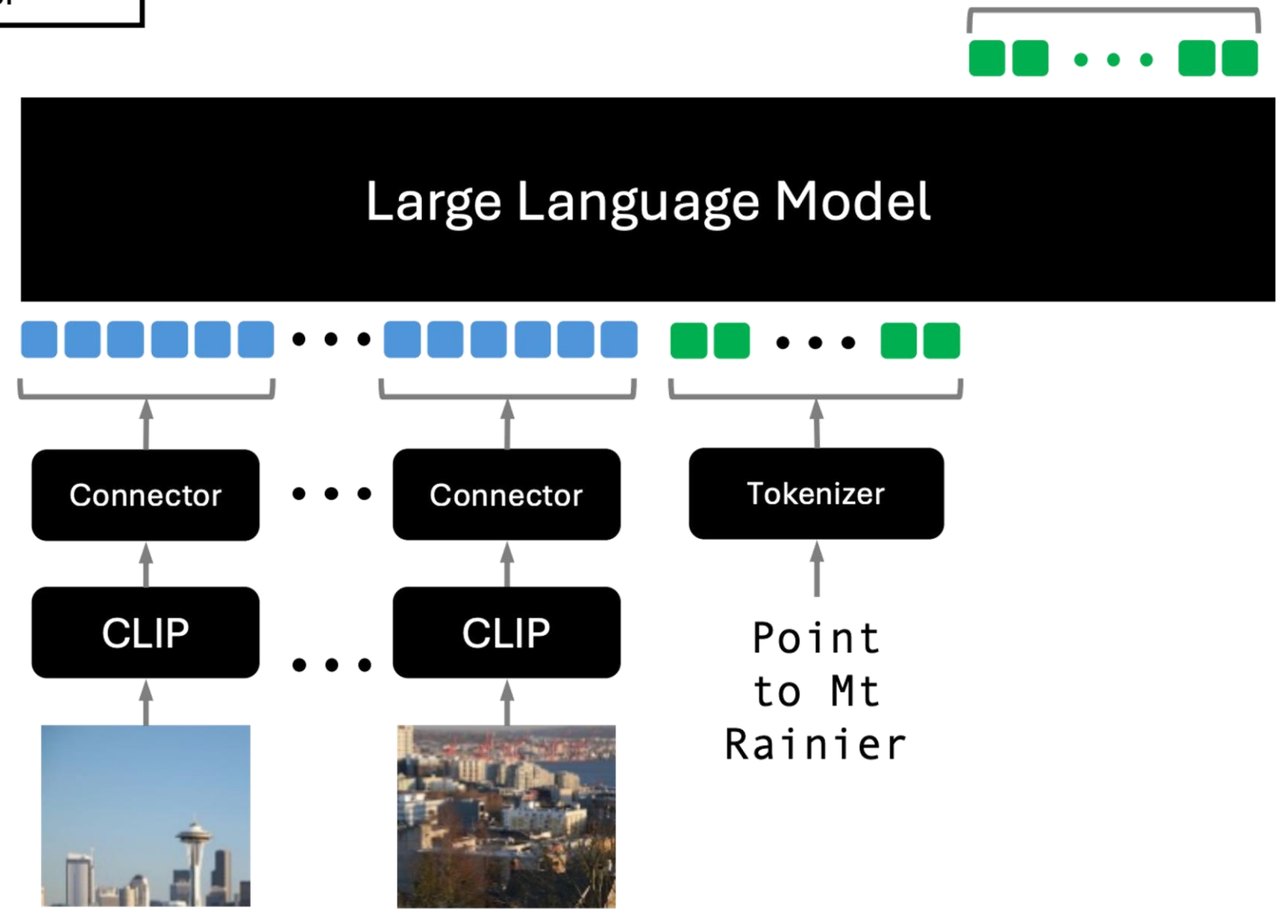
People don’t like to type
… but they love to talk

We ask annotators to speak for 60 to 90 seconds about an image

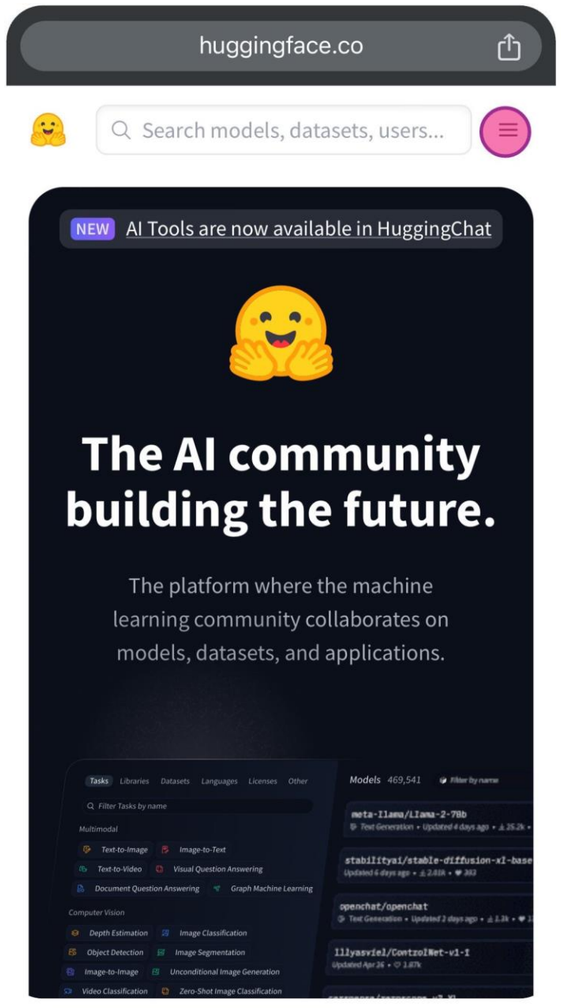
We automatically convert speech into text for pretraining



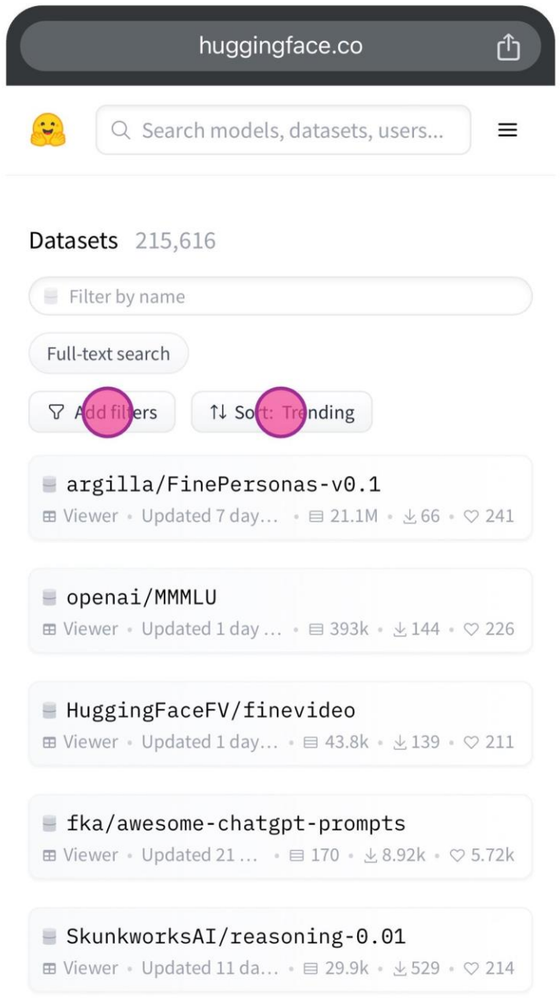
Start with off-the-self
Large Language Model
& Visual Encoder



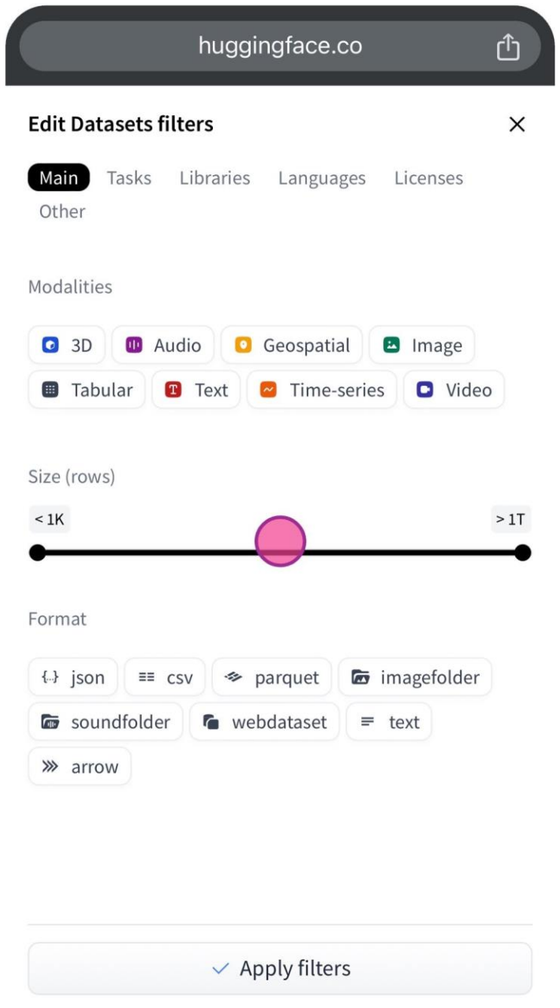
From perception To action



“Point to the menu”



“Point to where I can set search options”



“Point to where I can find mid size datasets”

Pointing to count, pointing to ground

Count the boats



- boats

Counting the **boats** shows a total of 35.

point_qa: What is the route of this bus?



- Route number and destination displayed on bus

The image shows a white bus with a green stripe along its side. **The bus is heading to Berkeley BART**, as indicated by the digital display on its front. Berkeley BART likely refers to the Berkeley station of the Bay Area Rapid Transit system, which is the main light rail and bus network serving the San Francisco Bay Area.

find person in this image and point me to that and count



person in this image

Counting the **person in this image** shows a total of 1.

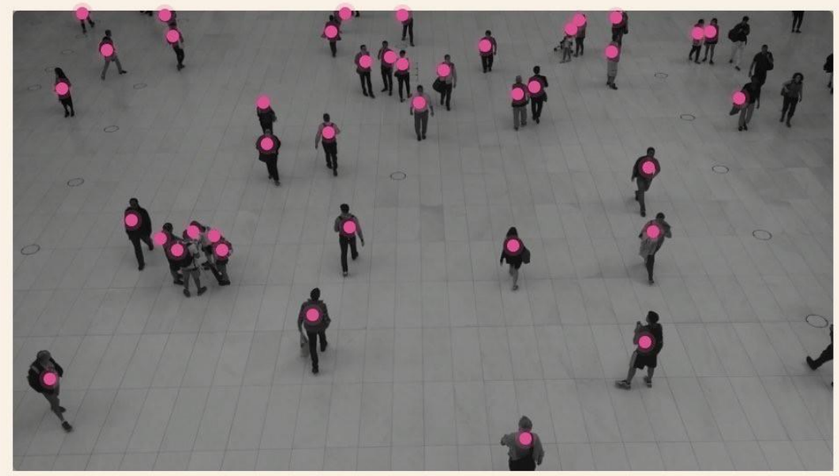
Detect and count red team players



red team players

Counting the **red team players** shows a total of 10.

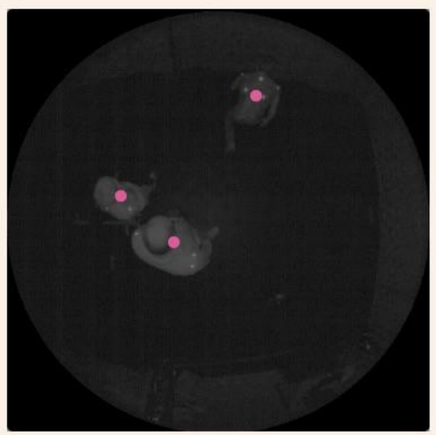
detect and count people



people

Counting the **people** shows a total of 40.

find person in this image and point me to that and count



person in this image

Counting the **person in this image** shows a total of 3.



cars on left lane

Counting the **cars on left lane** shows a total of 5.



cars on right lane

Counting the **cars on right lane** shows a total of 2.

Demo

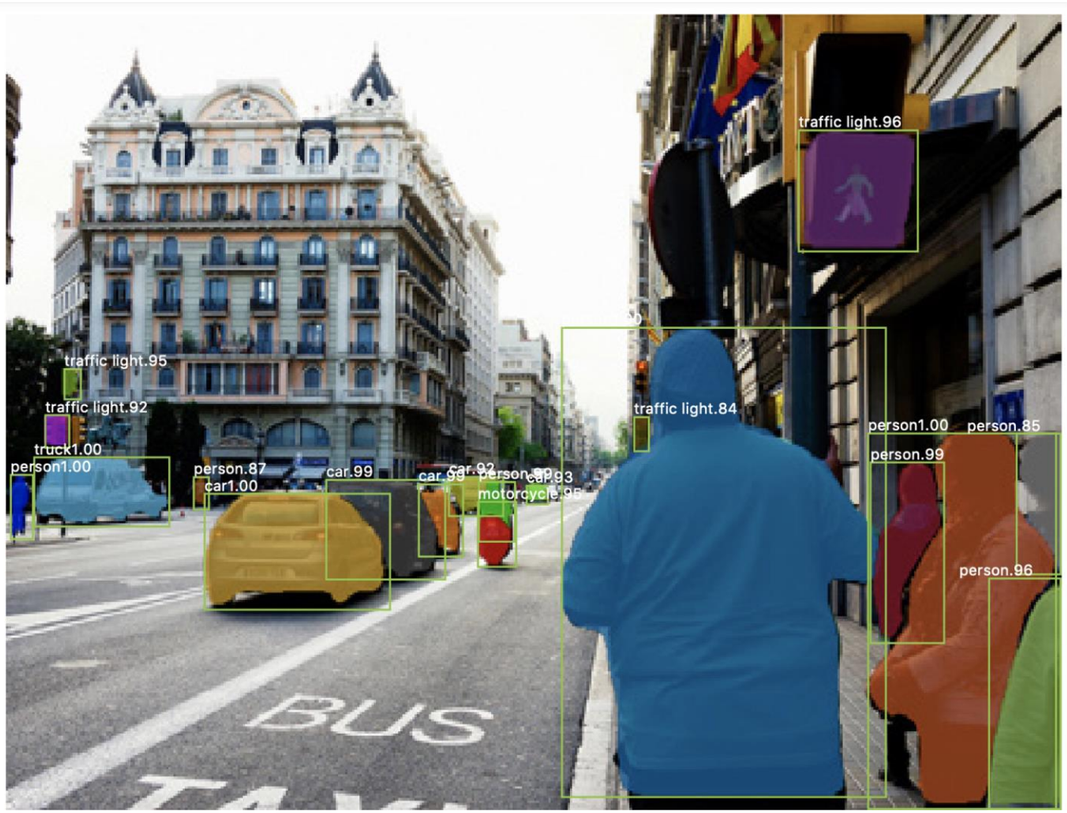
molmo.allenai.org

Foundation Models

Language	Classification	LM + Vision	And More!	Chaining
ELMo	**CLIP**	**LLaVA**	**Segment Anything**	**LMs + CLIP**
BERT	CoCa	Flamingo	Whisper	**Visual Programming**
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

Segment Anything Model (SAM)

What does it mean to have a segmentation foundation model?



Masking model trained on dataset of specific number of objects (80 in COCO)

Model outputs masks of all objects in that image that is one of the categories of interest

Images: He et al. Mask R-CNN. 2017

Segment Anything Model (SAM)

What does it mean to have a segmentation foundation model?



Masking model trained on a dataset of a huge number of categories

Model outputs mask of any objects that the user cares about

Images: Kirillov et al. Segment Anything. 2023.

Segment Anything Model (SAM)

What does it mean to have a segmentation foundation model?



Masking model trained on a dataset of a huge number of categories

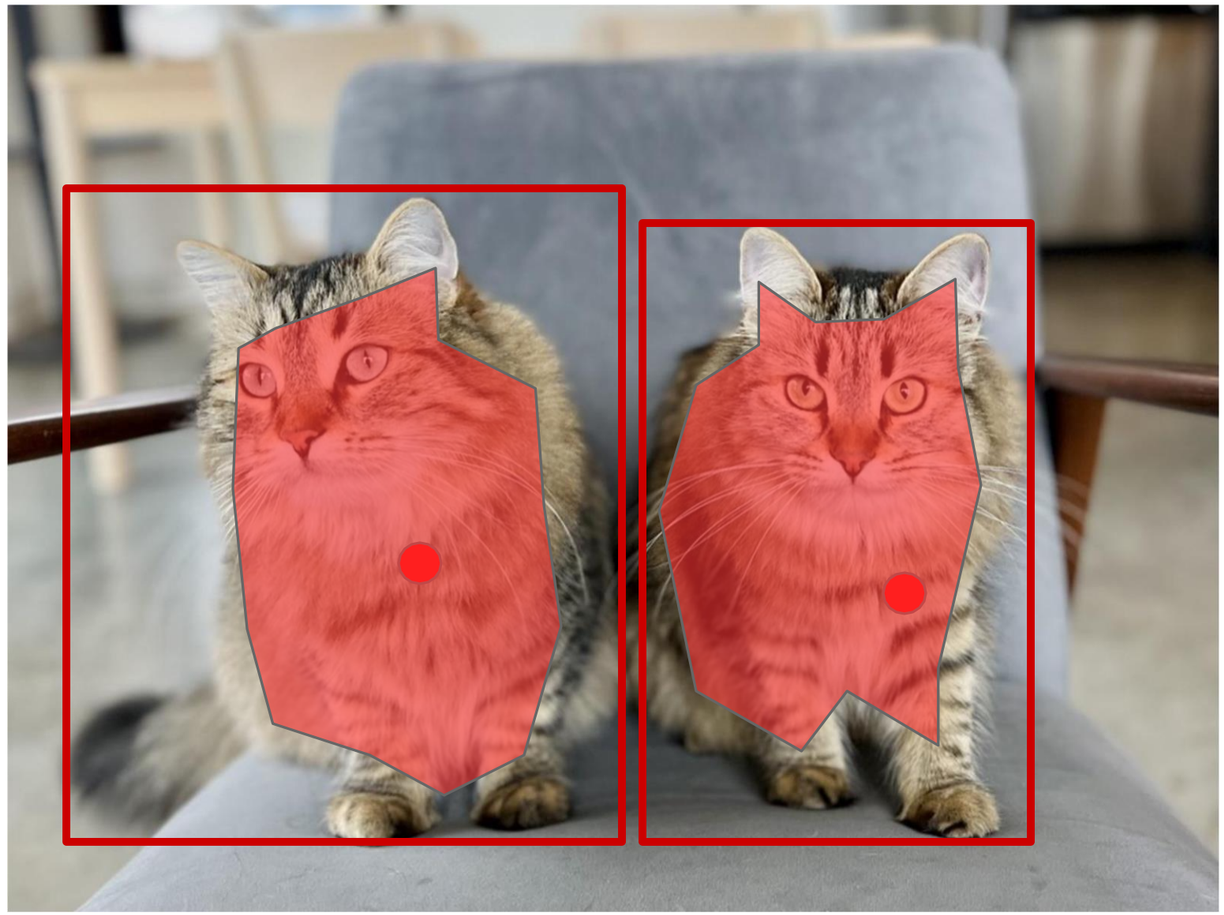
How to get this?

Model outputs mask of any objects that the user cares about

How to know this?

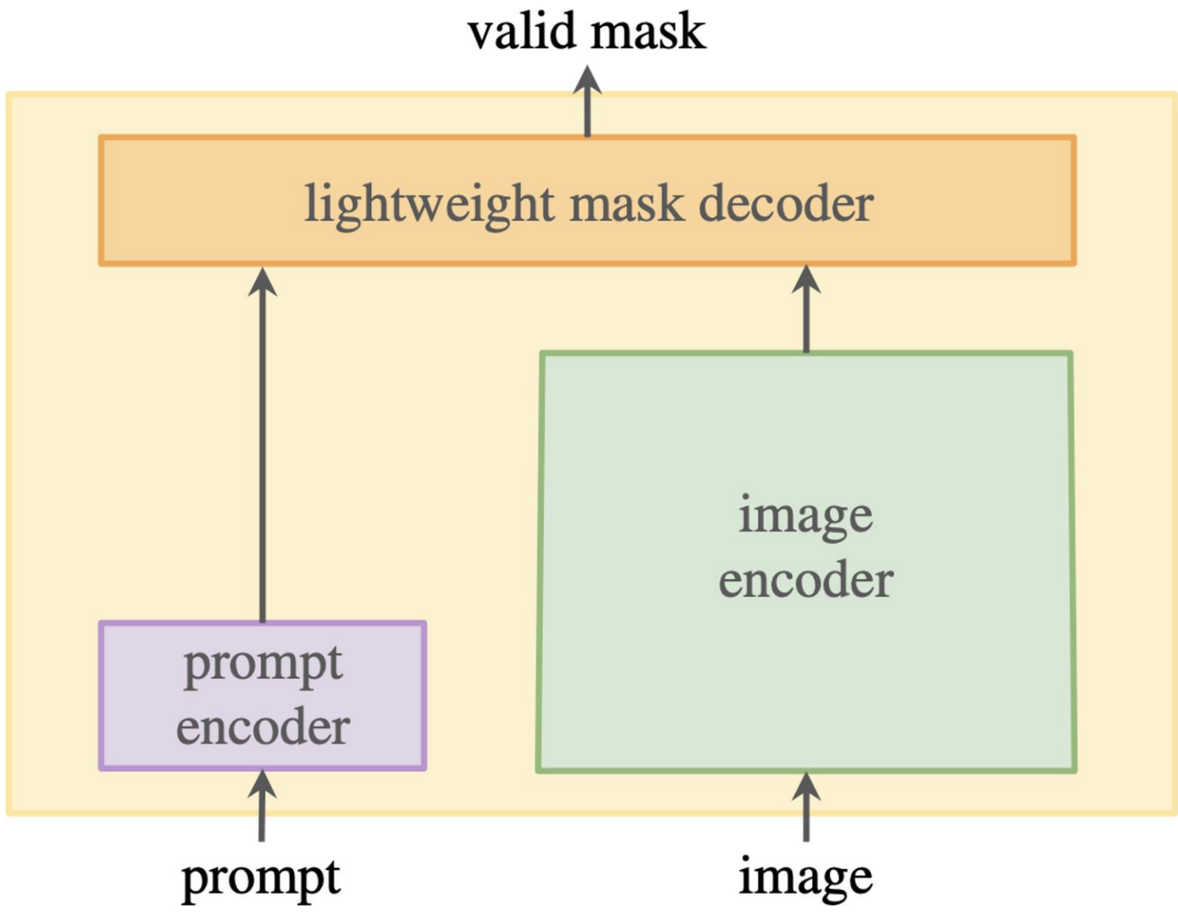
Images: Kirillov et al. Segment Anything. 2023.

How to know what to mask?



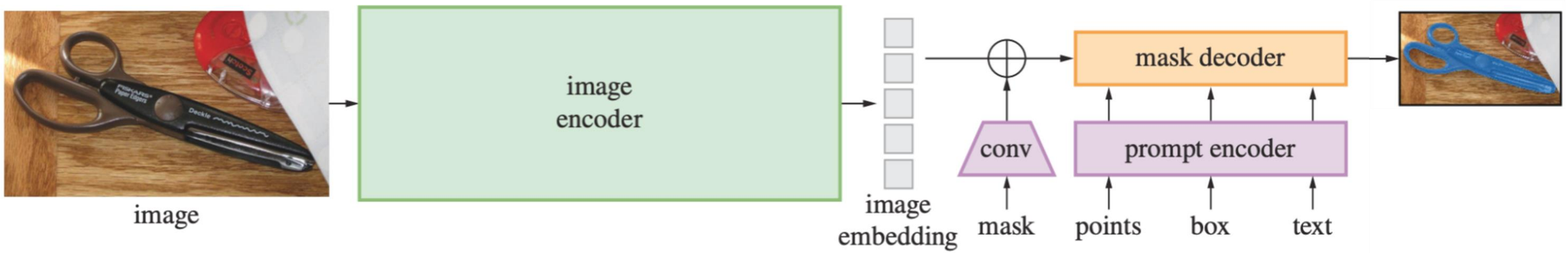
“Cats”

Basic SAM Architecture



Images: Kirillov et al. Segment Anything. 2023.

SAM Architecture



Images: Kirillov et al. Segment Anything. 2023.

Ambiguity in correct prompt



Images: Kirillov et al. Segment Anything. 2023.

Ambiguity in correct prompt



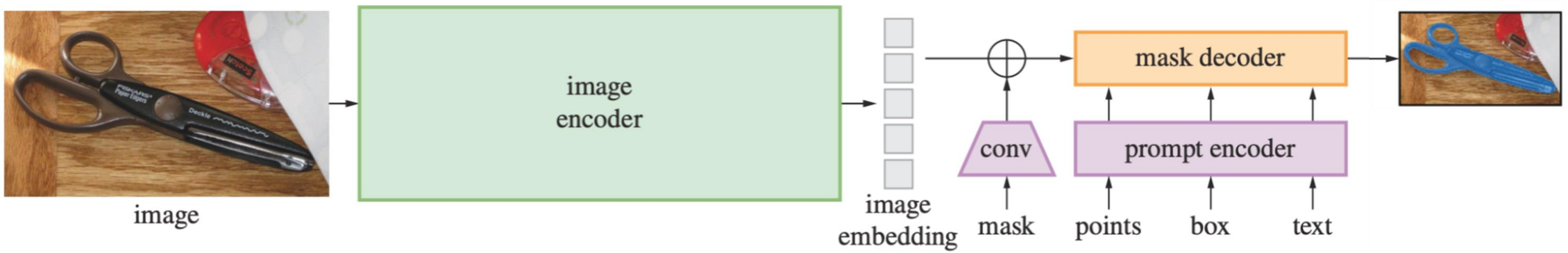






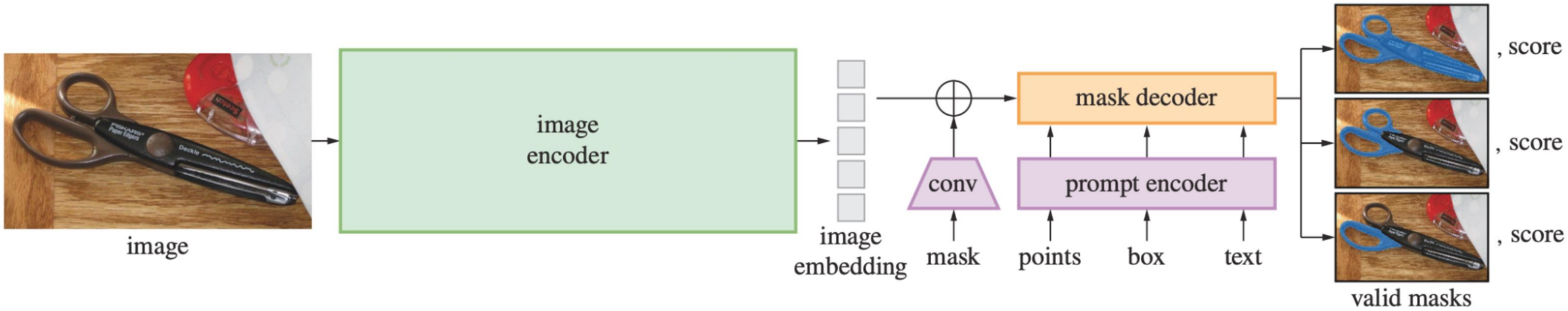
Images: Kirillov et al. Segment Anything. 2023.

SAM Architecture



Images: Kirillov et al. Segment Anything. 2023.

Basic SAM Architecture



1. Loss only calculated with respect to best mask
2. Model also trained to output confidence score for each mask

Images: Kirillov et al. Segment Anything. 2023.

Segment Anything Model (SAM)

What does it mean to have a segmentation foundation model?



Masking model trained on a dataset of a huge number of categories

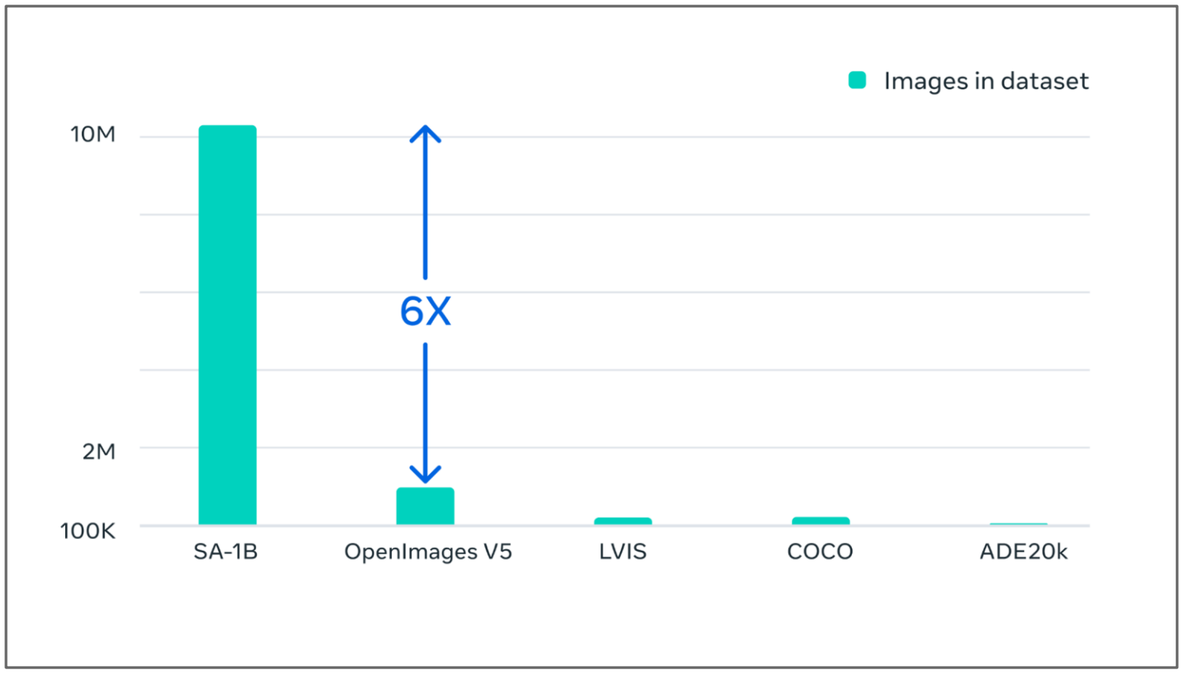
How to get this?

Model outputs mask of any objects that the user cares about

How to know this?

Images: Kirillov et al. Segment Anything. 2023.

Segment Anything Model (SAM)



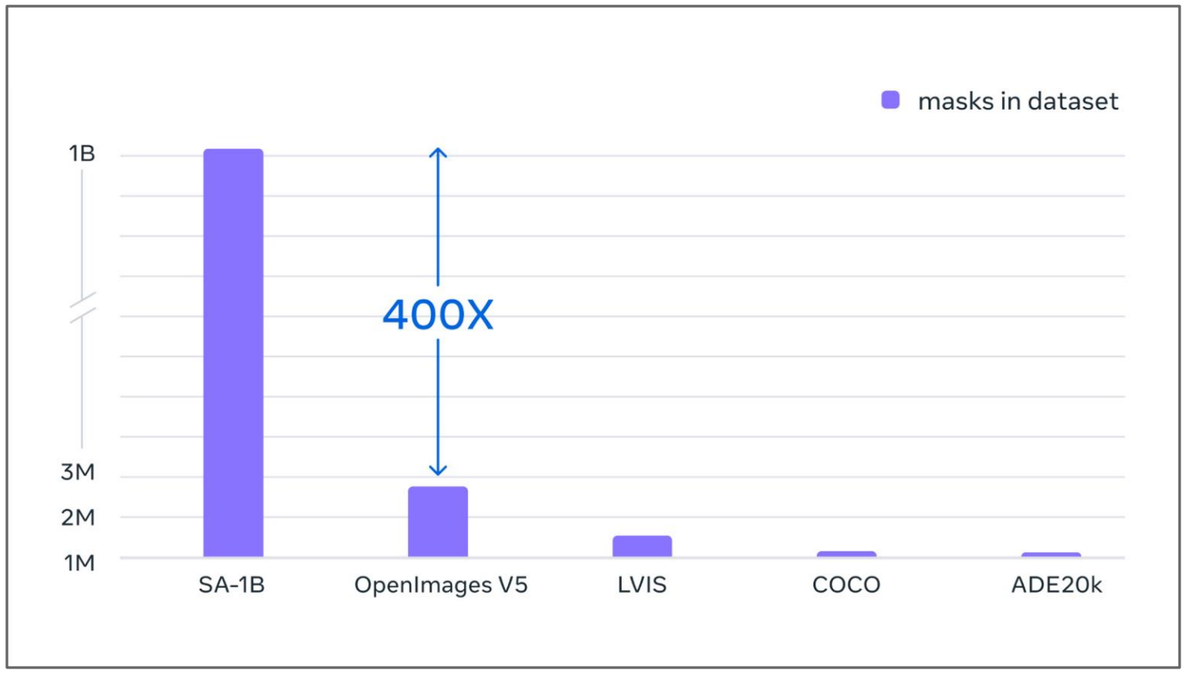
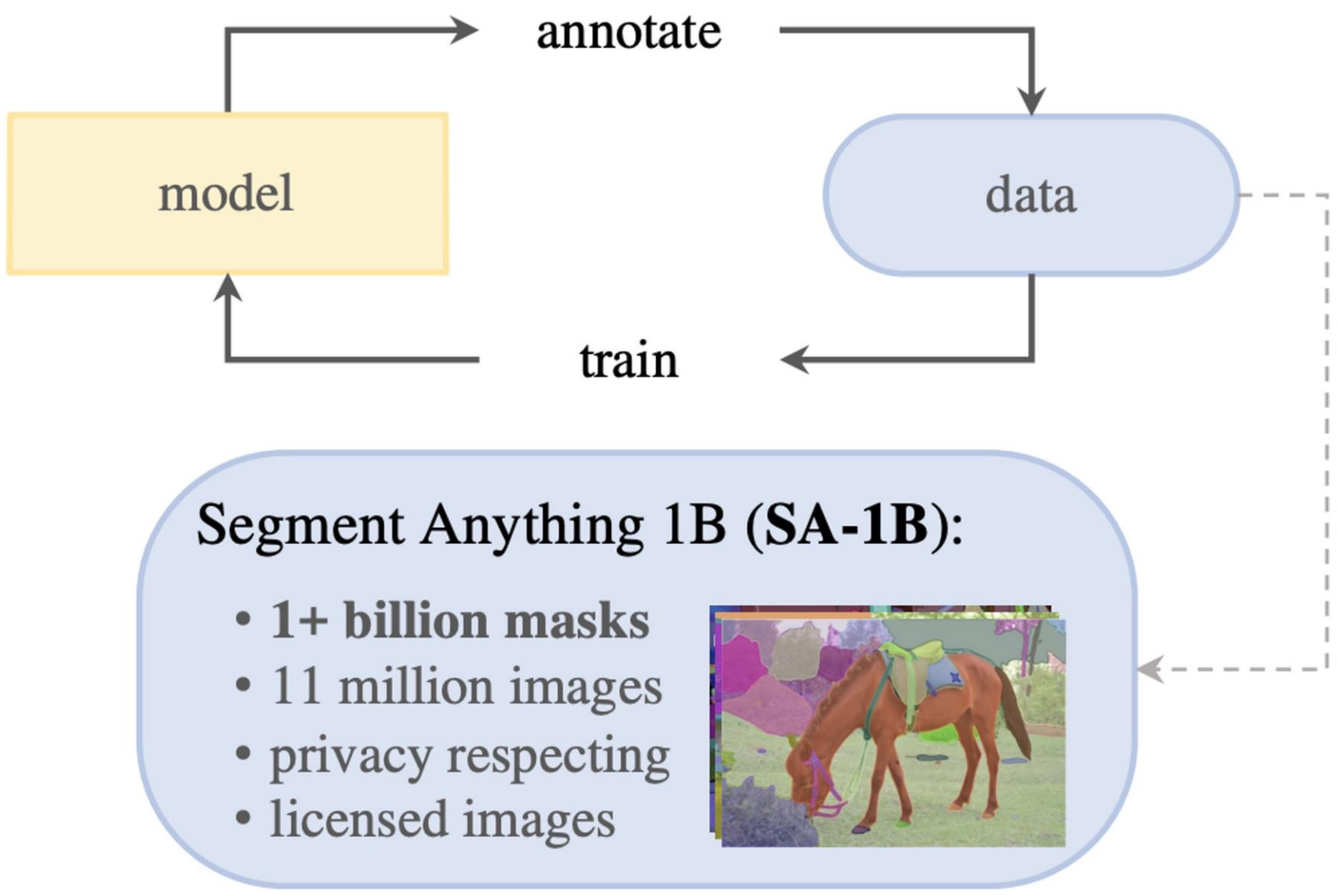


Image Source: https://segment-anything.com/

Segment Anything Model (SAM)



Images: Kirillov et al. Segment Anything. 2023.

SAM Results



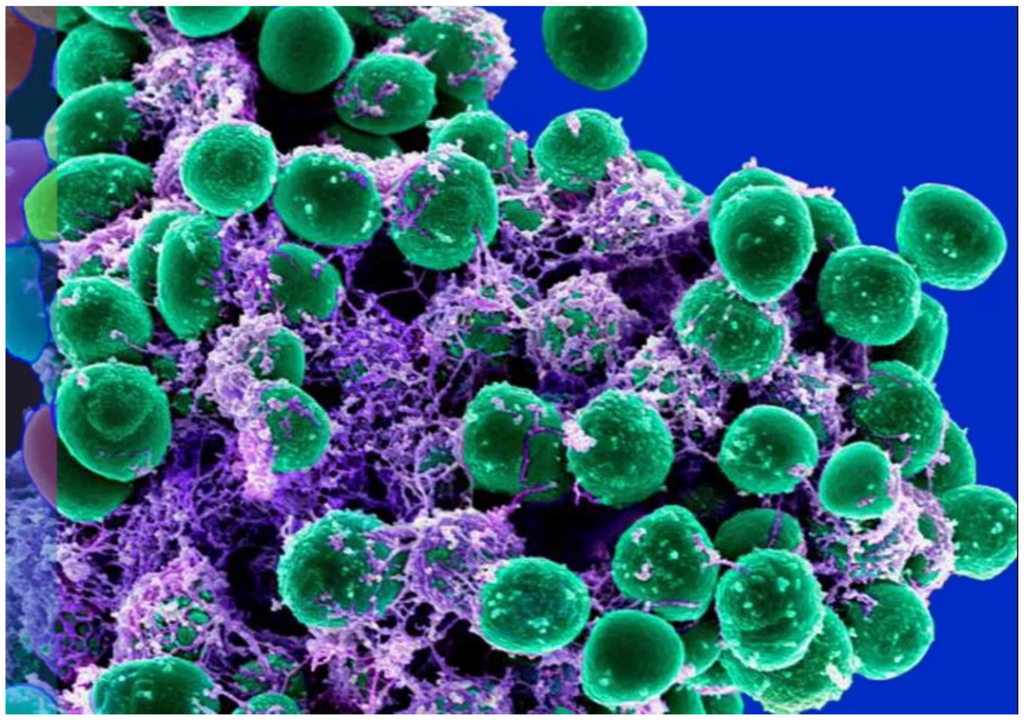
Image Source: Kirillov et al. Segment Anything. 2023

SAM Results



Image Source: Kirillov et al. Segment Anything. 2023

Zero-Shot with SAM



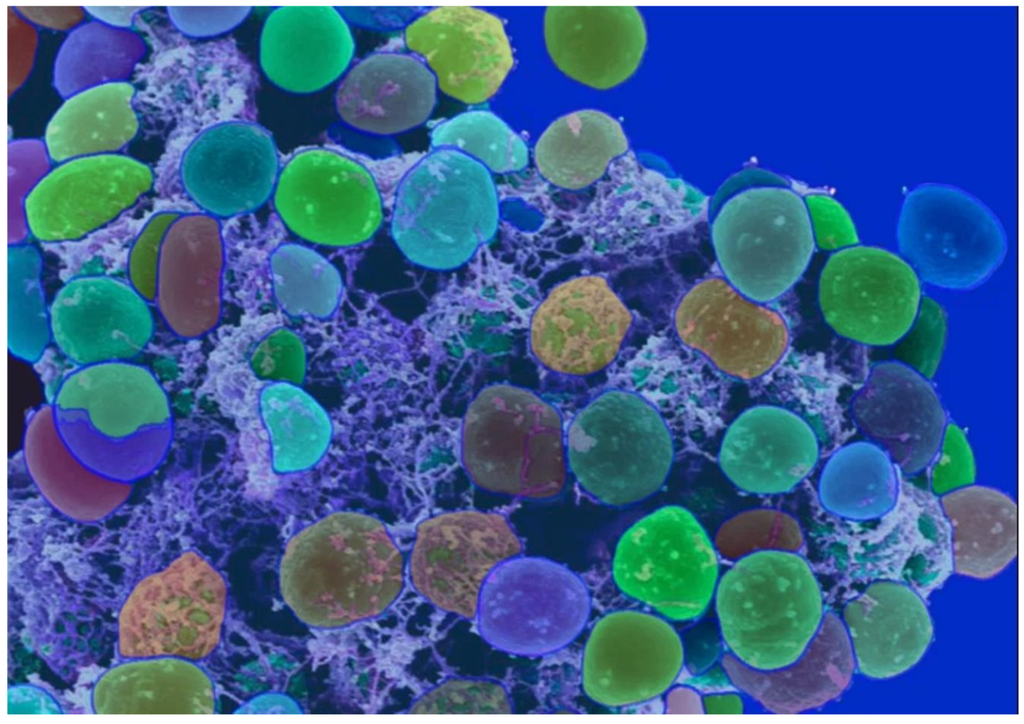


Image Source: https://segment-anything.com/

Zero-Shot with SAM





Image Source: https://segment-anything.com/

Chaining Molmo + SAM 2

can you point to the cricket bat





Future: Embodied AI for Navigation & Manipulation



P

Send

Points are shown in the image.

Foundation Models

Language	Classification	LM + Vision	And More!	Chaining
ELMo	**CLIP**	**LLaVA**	**Segment Anything**	**LMs + CLIP**
BERT	CoCa	Flamingo	Whisper	**Visual Programming**
GPT		GPT-4V	Dalle	
T5		Gemini	Stable Diffusion	
		Molmo	Imagen	

What happens when a model is asked to classify a concept it has never seen?

A photo of a marimba
A photo of a viaduct
A photo of a papillon
A photo of a lorikeet






Pratt et al “What does a platypus look like? Generating customized prompts for zero-shot image classification”. 2023.

”

Solution: chaining

1. Get an LLM to generate a description.
2. Classify using the description

"A **marimba** is a large wooden percussion instrument that looks like a xylophone."
"A **viaduct** is a bridge composed of several spans supported by piers or pillars."
"A **papillon** is a small, spaniel-type dog with a long, silky coat and fringed ears."
"A **lorikeet** is a small to medium-sized parrot with a brightly colored plumage."

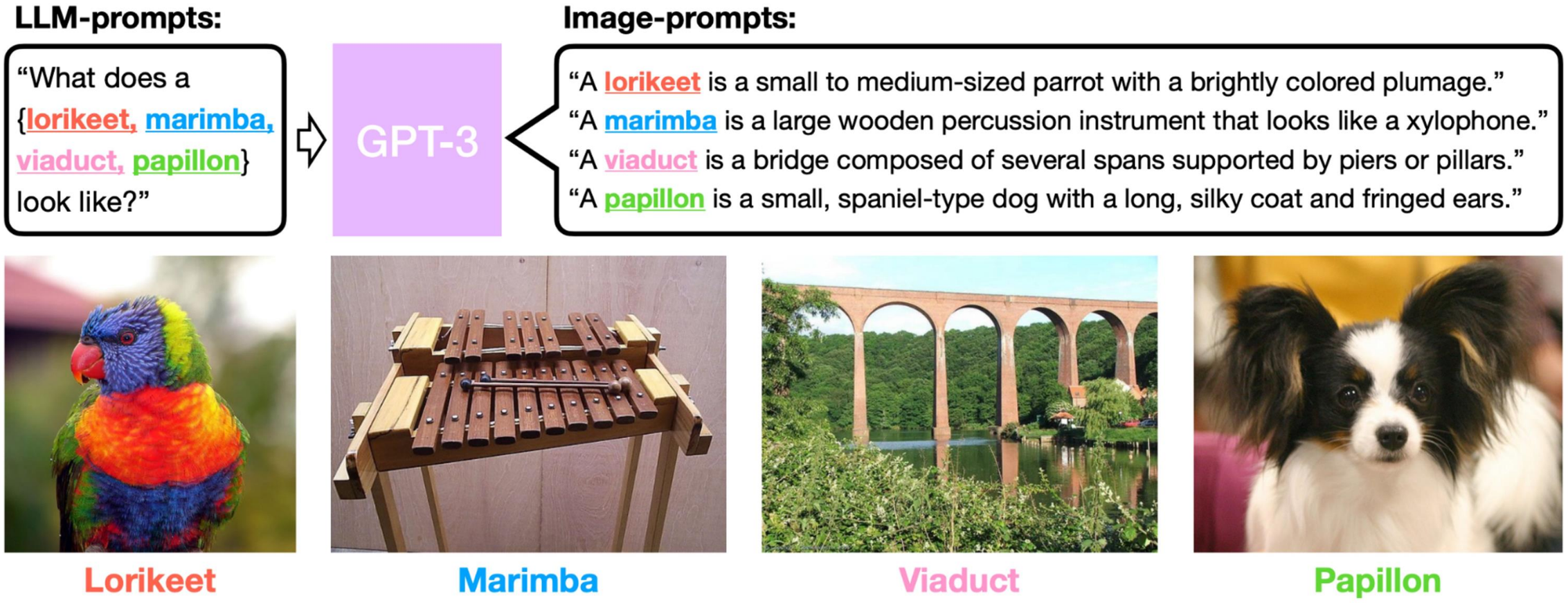





Pratt et al "What does a platypus look like? Generating customized prompts for zero-shot image classification". 2023.

"

CuPL (CUstomized Prompts via Language models)



Pratt et al "What does a platypus look like? Generating customized prompts for zero-shot image classification". 2023.

"

CuPL (CUstomized Prompts via Language models)

	ImageNet	DTD	Stanford Cars	SUN397	Food101	FGVC Aircraft	Oxford Pets	Caltech101	Flowers 102	UCF101	Kinetics-700	RESISC45	CIFAR-10	CIFAR-100	Birdsnap
std	75.54	55.20	77.53	69.31	93.08	32.88	93.33	93.24	78.53	77.45	60.07	71.10	95.59	78.26	50.43
# hw	80	8	8	2	1	2	1	34	1	48	28	18	18	18	1
CuPL (base)	76.19	58.90	76.49	72.74	93.33	36.69	93.37	93.45	78.83	77.74	60.24	68.96	95.81	78.47	51.11
Δ std	+0.65	+3.70	-1.04	+3.43	+0.25	+3.81	+0.04	+0.21	+0.30	+0.29	+0.17	-2.14	+0.22	+0.21	+0.63
# hw	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

Pratt et al “What does a platypus look like? Generating customized prompts for zero-shot image classification”. 2023.

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Can we generalize the idea of chaining to all vision tasks?

Many Visual Question Answering models which have been trained to do this type of task



Are there 3 people in the boat?

Gupta et al “Visual Programming: Compositional visual reasoning without training”. 2023.

”

VisProg (visual programming)

LEFT:



RIGHT:



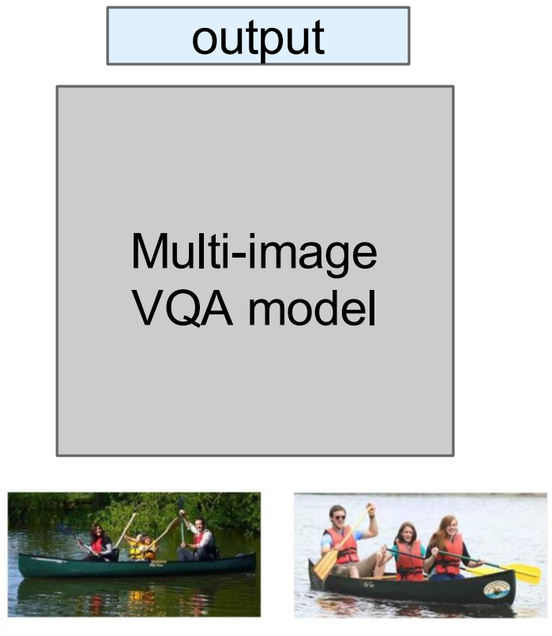
Statement: The left and right image contains a total of six people and two boats.

Gupta et al “Visual Programming: Compositional visual reasoning without training”. 2023.

”

VisProg (visual programming)

Train a new model for your task



Write a python script with the models you have

```
Class MyMultiImageVQA():

  Def ProcessIms():
    Ans1 = VQA(Image1)
    Ans2 = VQA(Image2)
    Return Ans1 + Ans2
```

General to 2 images now, but not beyond that

Gupta et al “Visual Programming: Compositional visual reasoning without training”. 2023.

”

VisProg (visual programming)

LEFT:



RIGHT:



Statement: The left and right image contains a total of six people and two boats.

GPT

```
Class MyMultiImageVQA():

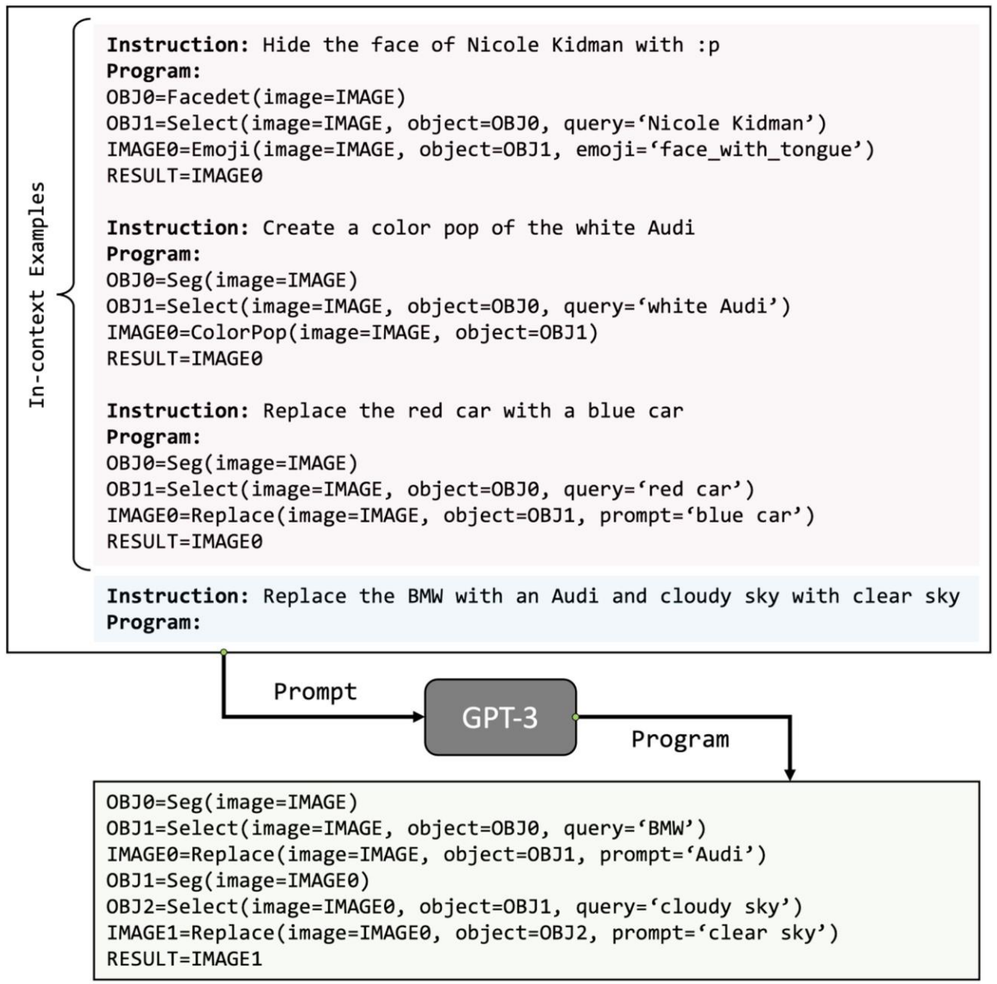
  Def ProcessIms():
    Ans1 = VQA(Image1)
    Ans2 = VQA(Image2)
    Return Ans1 + Ans2
```

→ False

Gupta et al “Visual Programming: Compositional visual reasoning without training”. 2023.

”

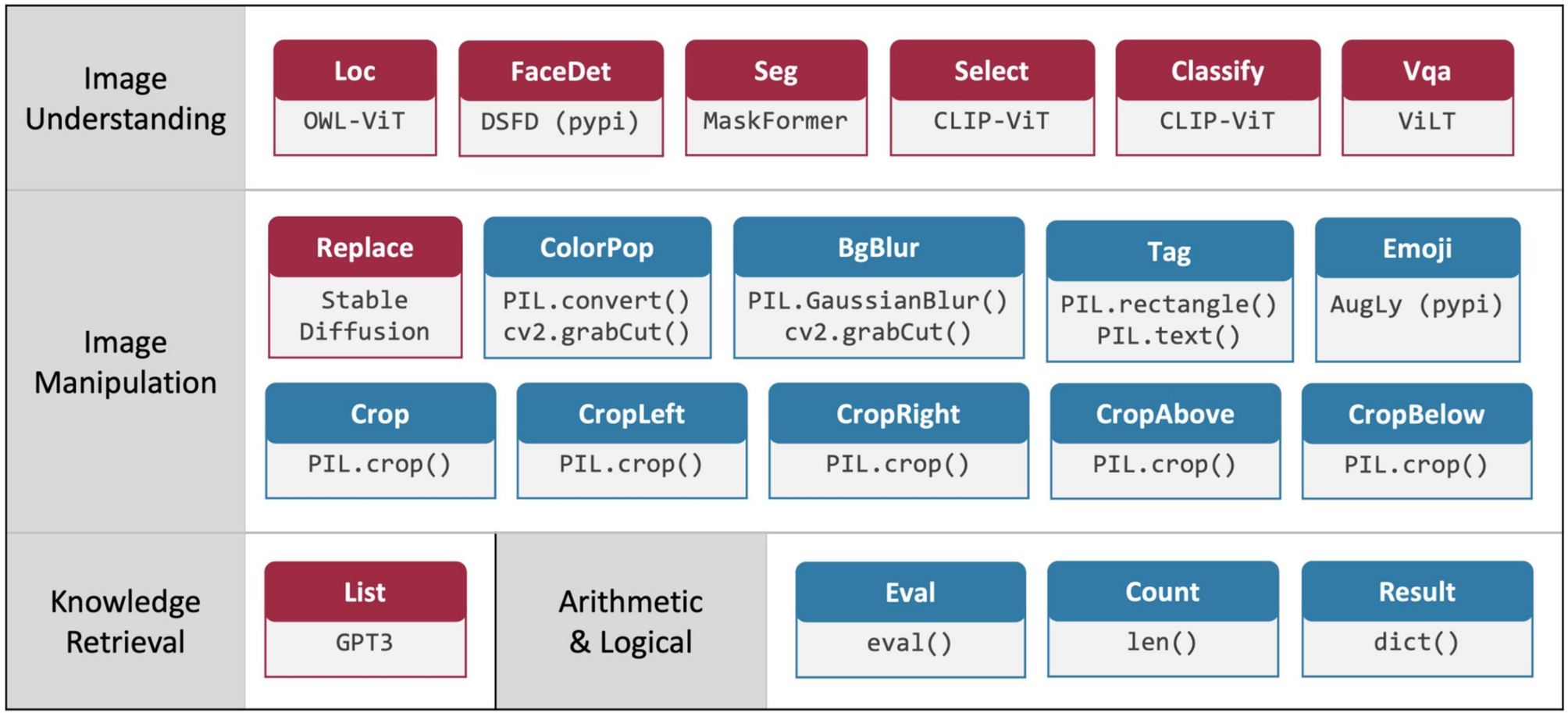
VisProg (visual programming)



Gupta et al "Visual Programming: Compositional visual reasoning without training". 2023.

"

VisProg (visual programming)



Gupta et al “Visual Programming: Compositional visual reasoning without training”. 2023.

”

VisProg (visual programming)

Natural Language Visual Reasoning

LEFT:



RIGHT:



Statement: The left and right image contains a total of six people and two boats.

Program:

```
ANSWER0=Vqa(image=LEFT, question='How many people are in the image?')
ANSWER1=Vqa(image=RIGHT, question='How many people are in the image?')
ANSWER2=Vqa(image=LEFT, question='How many boats are in the image?')
ANSWER3=Vqa(image=RIGHT, question='How many boats are in the image?')
ANSWER4=Eval('{ANSWER0} + {ANSWER1} == 6 and {ANSWER2} + {ANSWER3} == 2')
RESULT=ANSWER4
```

Prediction: False

Gupta et al "Visual Programming: Compositional visual reasoning without training". 2023.

"

VisProg (visual programming)

Factual Knowledge Object Tagging

IMAGE:



Prediction: IMAGE0



Instruction: Tag the 7 main characters on the TV show Big Bang Theory

Program:

```
OBJ0=FaceDet(image=IMAGE)
LIST0=List(query='main characters on the TV show Big Bang Theory', max=7)
OBJ1=Classify(image=IMAGE, object=OBJ0, categories=LIST0)
IMAGE0=Tag(image=IMAGE, object=OBJ1)
RESULT=IMAGE0
```

Gupta et al "Visual Programming: Compositional visual reasoning without training". 2023.

"

VisProg (visual programming)

IMAGE:



Prediction: IMAGE0



Instruction: Replace desert with lush green grass

Program:

```
OBJ0=Seg(image=IMAGE)
OBJ1=Select(image=IMAGE, object=OBJ0, query='desert', category=None)
IMAGE0=Replace(image=IMAGE, object=OBJ1, prompt='lush green grass')
RESULT=IMAGE0
```

Gupta et al "Visual Programming: Compositional visual reasoning without training". 2023.

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